

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

2008 BRAKES

Vehicle Dynamic Systems - Edge & MKX

SPECIFICATIONS

MATERIAL SPECIFICATIONS

Item	Specification	Fill Capacity
High Performance DOT 3 Motor Vehicle Brake Fluid PM-1-C (US); CPM-1-C (Canada)	WSS-M6C62-A or WSS-M6C65-A1	803-821 ml (1.69-1.73 pt)

TORQUE SPECIFICATIONS

Description	Nm	lb-ft	lb-in
ABS module screws	5	-	44
Brake caliper brake tube-to-hydraulic control unit (HCU) fittings	22	16	-
HCU bracket-to-frame bolts	25	18	-
HCU bracket-to-HCU bolt	15	11	-
Master cylinder brake tube-to-HCU fittings	27	20	-
Stability control sensor cluster nuts	9	-	80
Steering wheel rotation sensor bolts	1	-	9
Wheel speed sensor bolts	7	-	62

DESCRIPTION AND OPERATION

ANTI-LOCK CONTROL

ABS With Roll Stability Control (RSC) And Traction Control

The ABS with roll stability control (RSC) and traction control consists of the following components:

- Hydraulic control unit (HCU)
- ABS module (also controls the RSC system and traction control system)
- Stability/traction control switch
- Steering angle sensor module (SASM)
- Stability control sensor cluster (contains the accelerometer[s], roll rate sensor and yaw rate sensor)
- Front wheel speed sensors
- Front wheel speed sensor tone rings (integral to the halfshafts)
- Rear wheel speed sensors
- Rear wheel speed sensor tone rings are:
 - integral to the halfshaft on all wheel drive [AWD] vehicles.
 - integral to the wheel hub and bearing assembly on front wheel drive [FWD] vehicles.

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

- Brake pressure transducer (integral to the HCU)
- Brake fluid level switch

The ABS prevents wheel lock-up by monitoring the wheel speed sensors and actuating the valves in the HCU which modulates the brake pressure to the brake calipers. By preventing the wheels from locking up, the driver is able to maintain steering control and stop in the shortest possible distance under most conditions. The traction control system utilizes the same components as the ABS to prevent wheel spin and help maintain vehicle control during acceleration. The RSC system utilizes the same components as the ABS and the traction control systems and also monitors the steering wheel rotation sensor and the stability control sensor cluster to assist in stabilizing the vehicle during abrupt maneuvers.

The front wheel speed sensor tone rings are integral to the front halfshafts. Refer to **FRONT DRIVE HALFSHAFTS** article.

On AWD vehicles, the rear wheel speed sensor tone rings are integral to the rear halfshafts. Refer to **REAR DRIVE HALFSHAFTS** article.

On FWD vehicles, the rear wheel speed sensor tone rings are integral to the wheel hub and bearing assembly. Refer to **REAR SUSPENSION** article.

DIAGNOSTIC TESTS

ANTI-LOCK CONTROL

Special Tools

Illustration	Tool Name	Tool Number
 ST1137-A	73III Automotive Meter	105-R0057 or equivalent
 ST2574-A	Flex Probe Kit	105-R025B or equivalent
 ST3030-A	Rotunda Active Wheel Speed Sensor Tester	105-R0110
	Vehicle Communication Module (VCM) and Integrated Diagnostic	

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX



System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operation

Standard ABS

The standard ABS module manages anti-lock braking to maintain vehicle control during deceleration.

When the ignition switch is in the RUN position, the ABS module does a preliminary electrical check and, at approximately 20 km/h (12 mph), the hydraulic pump motor is turned on for approximately 1/2 second. During this time a buzzing or humming noise may be heard and a vibration maybe felt in the brake pedal. This is a normal condition. Any malfunction of the ABS disables the stability assist and the ABS warning indicator illuminates. However, the power-assist braking system functions normally.

The ABS module monitors and compares the rotational speed of each wheel. Wheel speeds are measured by the wheel speed sensor, which electrically senses each tooth of the sensor ring as it passes through the magnetic field of the sensor. When the ABS module detects an impending wheel lock, the ABS module commands the hydraulic pump motor on and commands the hydraulic control unit (HCU) to open and close the appropriate solenoid valves to modulate the brake pressure to the individual brake caliper(s). Once the affected wheel(s) return to the desired speed, the ABS module commands the HCU to return the solenoid valves to their normal position and normal base brake operation is restored.

In extreme cold conditions (temperature less than -25°C [-13°F]), the ABS module will briefly operate the hydraulic pump motor when the accelerator is pressed. This is done to warm the brake fluid.

Electronic Brake Distribution (EBD)

On initial application of the brake pedal, full pressure is applied to the rear brakes. The ABS module then uses wheel speed input to calculate an estimated rate of deceleration. Once vehicle deceleration exceeds a predetermined threshold, the ABS module closes the appropriate isolation valves in the Hydraulic Control Unit (HCU) to hold the rear brake pressure constant while allowing the front brake pressure to build. This creates a balanced braking condition between the front and rear wheels and minimizes the chance of rear wheel lockup during hard braking. As the vehicle decelerates, the valves are opened to increase the rear brake pressure in proportion to the front brake pressure.

A slight bump sensation may be felt in the brake pedal when Electronic Brake Distribution (EBD) is active.

If ABS is disabled due to DTCs being present in the ABS module, EBD will continue to function unless the DTCs are for wheel speed sensors or the HCU. When EBD is disabled, the ABS warning indicator and (if equipped) the Electronic Stability Control (ESC) warning indicator will be illuminated.

ABS with Traction Control, Electronic Stability Control (ESC) and Roll Stability Control (RSC)

The ABS module with traction control and roll stability control (RSC) performs the same functions as the standard ABS module. It also performs additional functions specific to traction control and RSC.

Traction Control

When the drive wheels lose traction and begin to spin, with vehicle speed under 100 km/h (62 mph), the ABS module commands the hydraulic pump motor on and commands the HCU to open and close the appropriate solenoid valves to modulate the brake pressure to the brake caliper(s) while simultaneously sending a request to the PCM over the high-speed controller area network (HS-CAN) bus to reduce engine torque to maintain vehicle traction. The PCM accomplishes this by minor incremental timing changes and fewer fuel injector pulses until the ABS module ends the request. The request ends when the driven wheel speed returns to the desired speed. After the vehicle speed exceeds 100 km/h (62 mph), the traction control is accomplished only through the PCM torque control.

Electronic Stability Control (ESC)

The electronic stability control (ESC) system constantly monitors the vehicle motion relative to the intended course. This is done by using sensors to compare the steering input and the yaw rate sensor with that of the actual vehicle motion. The ESC system monitors information from the throttle position (TP) sensor, the steering angle sensor module (SASM) (steering wheel angle and rate of change) and a yaw rate sensor (integral to the stability control sensor cluster) that measures changes in vehicle direction. If the ESC system determines from all these inputs that the vehicle is unable to travel in the intended direction, the system adjusts the brake torque at specific wheels in response to direct measurement of the vehicle motion and reduces engine torque to allow the vehicle to follow the intended course.

When the ESC system activates, any of the following can occur:

- A rumble or grinding sound much like ABS or traction control
- A small deceleration or a reduction in the acceleration of the vehicle
- The ESC indicator flashes
- If the driver's foot is on the brake pedal, a vibration will be felt in the pedal much like ABS

Roll Stability Control (RSC)

The roll stability control (RSC) system constantly monitors the vehicle motion relative to the intended course. This is done by using sensors to compare the steering input and brake application with that of the actual vehicle motion. The system does not activate when the vehicle is traveling in reverse; however, the standard ABS and traction control continue to function as usual. The RSC system monitors information from the wheel speed sensors, the TP sensor, the steering angle sensor module (SASM) (steering wheel angle and rate of change) and a roll-rate sensor (integral to the stability control sensor cluster) that measures changes in vehicle motion. If the RSC system determines from all these inputs that conditions exist for a potential roll-over event, the system adjusts the brake torque at specific wheels in response to direct measurement of the vehicle roll motion and reduces engine torque to make the vehicle more stable. By adjusting brake torque, the system can reduce the cornering forces and, therefore, the total roll moment acting on the vehicle.

When the RSC system activates, any of the following can occur:

- A rumble or grinding sound much like ABS or traction control
- A small deceleration or a reduction in the acceleration of the vehicle
- The RSC indicator flashes
- If the driver's foot is on the brake pedal, a vibration will be felt in the pedal much like ABS
- If the event is severe enough and the driver's foot is not on the brake, the brake pedal will move to apply higher brake forces. A whooshing sound can also be heard from under the instrument panel in an event this severe

Some drivers may notice a slight movement of the brake pedal when the system checks itself. The brake pedal moves when an active test of the brake booster is run. During this test a small amount of pressure is generated at the master cylinder, but no pressure is generated in the brake calipers.

Stability/Traction Control Switch

The stability/traction control switch is hard wired to the instrument cluster module. The instrument cluster module communicates the stability/traction control switch status to the ABS module via the HS-CAN bus.

The stability/traction control switch allows the driver to control use of the stability control/traction assist system. This is independent of the ABS function, which cannot be switched OFF by the driver. The stability control/traction assist system status is indicated by a light in the stability/traction control switch. When the stability/traction control switch is set to OFF, the system is disabled below 60 km/h (37 mph). The system will automatically restore to full functionality every time the vehicle speed exceeds 70 km/h (43 mph) or when the ignition switch is turned to the OFF position and back to the ON position. However, the stability/traction control switch OFF indicator remains illuminated until the switch is pressed. The ABS function continues to work as designed unless the yellow ABS warning indicator is also illuminated. Normal braking function always occurs, unless the red brake warning indicator is illuminated.

Stability Control Sensor Cluster

The stability control sensor cluster consists of the yaw rate sensor, roll-rate sensor, lateral accelerometer and longitudinal accelerometer. The lateral accelerometer measures the acceleration which corresponds to the force involved when the vehicle slides sideways. This acceleration has 2 forms. The first is the centrifugal acceleration which is generated when the vehicle travels around in a circle. The second is the acceleration due to gravity. On level ground there is no contribution from this acceleration. However, if the vehicle is parked sideways on a bank or incline, the sensor measures some lateral acceleration due to gravity even though the vehicle is not moving.

The longitudinal accelerometer measures the acceleration corresponding to the force involved when the vehicle moves forward and rearward in the horizontal plane, along the centerline of the front and rear wheels.

The yaw rate sensor measures the relative vehicle motion about the vertical axis through its center of gravity.

The roll-rate sensor measures the rate of rotation along the front to rear horizontal axis.

Steering Angle Sensor Module (SASM)

The steering angle sensor module (SASM) measures the rate of rotation of the steering wheel by monitoring the steering wheel rotation ring as it passes through the sensor gap. The SASM uses the HS-CAN bus to transmit information to the ABS module about whether the steering wheel is being turned left or right and how far it is being turned.

ABS Module Configuration

A new ABS module must be configured after installation. Refer to Programmable Module Installation in **MODULE CONFIGURATION** article. If installing a new ABS module with RSC, the module must also be calibrated.

ABS Module Calibration

NOTE: **The installation of a new stability control sensor cluster does not require ABS module calibration.**

The ABS module must be calibrated if any of the following is occurring or have occurred:

- A new ABS module has been installed.
- A new HCU has been installed.
- A new steering wheel rotation sensor has been installed.
- The ABS warning indicator in the instrument cluster flashes once every 2 seconds.

Follow the scan tool directions for ABS module calibration. The ABS module should be configured before it is calibrated. Calibration is required for the ABS module to learn the "at rest" or "zero" position of the stability control components. If a DTC is set for any stability control component, the DTC must be cleared before carrying out the calibration procedure. The ESC warning indicator will flash after clearing the DTCs that are associated with the stability control system. If a DTC is retrieved after calibration, refer to the **ABS MODULE DTC CHART** .

Stability Control Sensor Cluster Calibration

When a new stability control sensor cluster is installed, the cluster must be calibrated. Follow the scan tool instructions for the calibration procedure. When carrying out the cluster calibration, the vehicle must be on a level surface, must not be moving and the ignition switch must be in the RUN position.

Inspection And Verification

1. Verify the customer concern by applying the brakes under different conditions.
2. Verify the base brake system operates correctly by pressing and releasing the brake pedal with the ignition switch in the OFF position. If the stoplamps do not operate correctly, refer to **EXTERIOR LIGHTING** article. If the stoplamps operate correctly, proceed to the next step.
3. Verify the base brake system operates correctly. If any concerns are present, address them before continuing with any ABS concerns. Refer to **BRAKE SYSTEM - GENERAL INFORMATION** article. If the base brake system operates correctly, proceed to the next step.

4. Visually inspect for obvious signs of mechanical or electrical damage.

VISUAL INSPECTION CHART

Mechanical	Electrical
• Base brake system • Brake booster vacuum hose • Brake fluid level • Hydraulic control unit (HCU) • Steering components, suspension components and tire size • Steering angle sensor module (SASM) • Tire pressure • Tire size or mismatched tires • Wheel and tire sizes	• ABS module • Battery junction box (BJB) fuse(s): ○ 15 (40A) ○ 50 (10A) ○ 57 (40A) ○ 71 (10A) • Brake fluid level switch • Brake on/off (BOO) switch • Smart junction box (SJB) fuse(s): ○ 34 (5A) ○ 35 (10A) • Wiring, terminals or connectors • Stability control sensor cluster • Stability/traction control switch • Wheel speed sensor

5. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

NOTE: Make sure to use the latest scan tool software release.

6. If the cause is not visually evident, connect the scan tool to the data link connector (DLC).

NOTE: The vehicle communication module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM.

7. If the scan tool does not communicate with the VCM:

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM.
- refer to **MODULE COMMUNICATIONS NETWORK** article, No Power To The Scan Tool, to diagnose no communication with the scan tool.

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

8. If the scan tool does not communicate with the vehicle:
 - verify the ignition key is in the ON position.
 - verify the scan tool operation with a known good vehicle.
 - refer to **MODULE COMMUNICATIONS NETWORK** article to diagnose no response from the PCM.
9. Carry out the network test.
 - If the scan tool responds with no communication for one or more modules, refer to **MODULE COMMUNICATIONS NETWORK** article.
 - If the network test passes, retrieve and record continuous memory DTCs.
10. Clear the continuous DTCs and carry out the self-test diagnostics for the ABS module, the instrument cluster and the steering angle sensor module.
11. If the DTCs retrieved are related to the concern, go to the **ABS MODULE DTC CHART** , the Steering Angle Sensor Module DTC Chart and/or the Instrument Cluster Module DTC Chart. For all other DTCs, refer to the Master DTC Chart in **MULTIFUNCTION ELECTRONIC MODULES** article.
12. If no DTCs related to the concern are retrieved, go to **Symptom Chart**.

ABS MODULE DTC CHART

DTC	Description	Source	Action
B1317	Battery Voltage High	ABS Module	NOTE: Battery voltage above 16 volts will cause DTC B1317 to set. Verify battery and charging system condition prior to diagnosing this DTC. Go to Pinpoint Test A .
B1318	Battery Voltage Low	ABS Module	NOTE: Battery voltage below 10 volts will cause DTC B1318 to set. Verify battery and charging system condition prior to diagnosing this DTC. Go to Pinpoint Test A .
B1342	ECU is Faulted	ABS Module	NOTE: If other DTCs are present, REPAIR them before installing a new component. CLEAR the DTCs. CARRY OUT the ABS module self-test. RETRIEVE the DTCs. If DTC B1342 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

			the self-test with the brake pedal not applied.
B211B	Satellite ECU is Faulted	ABS Module	CHECK for water intrusion into stability control sensor cluster connector C3037. REPAIR as necessary. If OK, INSTALL a new sensor cluster. REFER to <u>Stability Control Sensor Cluster</u> . CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
B2477	Module Configuration Failure	ABS Module	CONFIGURE the ABS module. REFER to <u>MODULE CONFIGURATION</u> article. CLEAR the DTCs. REPEAT the self-test. If DTC B2477 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . REPEAT the self-test.
C1095	ABS Hydraulic Pump Motor Circuit Failure	ABS Module	Go to Pinpoint Test B .
C1096	ABS Hydraulic Pump Motor Circuit Open	ABS Module	Go to Pinpoint Test B .
C1115	ABS Power Relay Output Short Circuit to Battery	ABS Module	NOTE: If other DTCs are present, REPAIR them before installing a new component. CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs.If DTC C1115 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u>. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1140	Hydraulic Base Brake Failure	ABS Module	The ABS module receives brake fluid level information from the instrument cluster over the high-speed controller area network (HS-CAN) bus. The instrument cluster receives the brake fluid information from the smart junction box (SJB) over the medium-speed controller area network (MS-CAN) bus. This DTC indicates that a low fluid level condition exists or there is an issue with the brake fluid level switch, wiring, instrument cluster or HS-CAN bus. If there are other HS-CAN bus DTCs present, REFER to MODULE

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

			<u>COMMUNICATIONS NETWORK</u> article to diagnose the HS-CAN bus. If there are no HS-CAN bus DTCs present, VERIFY that the brake fluid is at the correct level and ADJUST as necessary. If the brake fluid level is OK, REFER to <u>INSTRUMENT CLUSTER (IC), MESSAGE CENTER & WARNING CHIMES</u> article to diagnose the red brake warning indicator and the brake fluid level switch.
C1145	Speed Wheel Sensor RF Input Circuit Failure	ABS Module	Go to Pinpoint Test C .
C1155	Speed Wheel Sensor LF Input Circuit Failure	ABS Module	Go to Pinpoint Test C .
C1165	Speed Wheel Sensor RR Input Circuit Failure	ABS Module	Go to Pinpoint Test C .
C1175	Speed Wheel Sensor LR Input Circuit Failure	ABS Module	Go to Pinpoint Test C .
C1185	ABS Power Relay Output Circuit Failure	ABS Module	NOTE: If other DTCs are present, REPAIR them before installing a new component. CLEAR all DTCs. RETRIEVE DTCs. If DTC C1185 is retrieve again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u>. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1194	ABS Outlet Valve Coil LF Circuit Failure	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If C1194 is retrieved again along with one or more other ABS valve coil DTCs, go to Pinpoint Test A . If only DTC C1194 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u>. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
			CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

C1198	ABS Inlet Valve Coil LF Circuit Failure	ABS Module	at least one ABS stop. RETRIEVE the DTCs. If C1198 is retrieved again along with one or more other ABS valve coil DTCs, go to Pinpoint Test A . If only DTC C1198 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1210	ABS Outlet Valve Coil RF Circuit Failure	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If C1210 is retrieved again along with one or more other ABS valve coil DTCs, go to Pinpoint Test A . If only DTC C1210 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1214	ABS Inlet Valve Coil RF Circuit Failure	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If C1214 is retrieved again along with one or more other ABS valve coil DTCs, go to Pinpoint Test A . If only DTC C1214 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1222	Speed Wheel Mismatch	ABS Module	Go to Pinpoint Test D .
C1233	Speed Wheel LF Input Signal Missing	ABS Module	Go to Pinpoint Test D .
C1234	Speed Wheel RF Input Signal Missing	ABS Module	Go to Pinpoint Test D .
C1235	Speed Wheel RR Input Signal Missing	ABS Module	Go to Pinpoint Test D .
C1236	Speed Wheel LR Input Signal Missing	ABS Module	Go to Pinpoint Test D .

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

C1242	ABS Outlet Valve Coil LR Circuit Failure	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If C1242 is retrieved again along with one or more other ABS valve coil DTCs, go to Pinpoint Test A. If only DTC C1242 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u>. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1246	ABS Outlet Valve Coil RR Circuit Failure	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If C1246 is retrieved again along with one or more other ABS valve coil DTCs, go to Pinpoint Test A. If only DTC C1246 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u>. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1250	ABS Inlet Valve Coil LR Circuit Failure	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If C1250 is retrieved again along with one or more other ABS valve coil DTCs, go to Pinpoint Test A. If only DTC C1250 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u>. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1254	ABS Inlet Valve Coil RR Circuit Failure	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If C1254 is retrieved again along with one or more other ABS valve coil DTCs, go to Pinpoint Test A. If only DTC C1254 is retrieved again, INSTALL a new ABS module. REFER to

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

			Anti-Lock Brake System (ABS) Module. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1278	Steering Wheel Angle 1 and 2 Signal Faulted	ABS Module	Go to Pinpoint Test E .
C1279	Yaw Rate Sensor Circuit Failure	ABS Module	Go to Pinpoint Test F .
C1280	Yaw Rate Sensor Signal Fault	ABS Module	Go to Pinpoint Test F .
C1281	Lateral Accelerometer Circuit Failure	ABS Module	Go to Pinpoint Test F .
C1282	Lateral Accelerometer Signal Fault	ABS Module	Go to Pinpoint Test F .
C1288	Brake Pressure Transducer Main/Primary Input Circuit Failure	ABS Module	Go to Pinpoint Test G .
C1329	ABS Outlet Valve Coil RF Circuit Excessive Temperature	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If the DTC is retrieved again, INSTALL a new HCU. REFER to Hydraulic Control Unit (HCU) . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1330	ABS Outlet Valve Coil LR Circuit Excessive Temperature	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If the DTC is retrieved again, INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1331	ABS Outlet Valve Coil RR Circuit Excessive Temperature	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If the DTC is retrieved again, INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

			the self-test with the brake pedal not applied.
C1332	ABS Outlet Valve Coil LF Circuit Excessive Temperature	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If the DTC is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1333	ABS Inlet Valve Coil LR Circuit Excessive Temperature	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If the DTC is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1334	ABS Inlet Valve Coil LF Circuit Excessive Temperature	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If the DTC is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1335	ABS Inlet Valve Coil RF Circuit Excessive Temperature	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If the DTC is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1336	ABS Inlet Valve Coil RR Circuit Excessive	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If the DTC is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> .

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

	Temperature		CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1404	Traction Control Valve Rear Circuit Failure	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If C1404 is retrieved again along with one or more other ABS valve coil DTCs, go to Pinpoint Test A . If only DTC C1404 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1410	Traction Control Valve LF Circuit Failure	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If C1410 is retrieved again along with one or more other ABS valve coil DTCs, go to Pinpoint Test A . If only DTC C1410 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1446	Brake Switch Circuit Failure	ABS Module	Go to Pinpoint Test H .
C1516	Roll Rate Sensor Signal Fault	ABS Module	Go to Pinpoint Test F .
C1517	Roll Rate Sensor Circuit Fault	ABS Module	Go to Pinpoint Test F .
C1527	Traction Control Rear Valve Circuit Excessive Temperature	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If the DTC is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
			CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs.

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

C1528	Traction Control LF Valve Circuit Excessive Temperature	ABS Module	If the DTC is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u>. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1530	Dynamic Stability Control RF Valve Circuit Excessive Temperature	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If the DTC is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u>. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1531	Dynamic Stability Control LF Valve Circuit Excessive Temperature	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If the DTC is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u>. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1730	Reference Voltage Out of Range (+5 V)	ABS Module	If any other DTCs are present, REPAIR those DTCs first. CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE DTCs. If the DTC is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u>. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1957	Dynamic Stability Control Valve RF Circuit Failure	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If C1957 is retrieved again along with one or more other ABS valve coil DTCs, go to Pinpoint Test A . If only DTC C1957 is retrieved again, INSTALL a new ABS module. REFER to

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

			Anti-Lock Brake System (ABS) Module. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1958	Stability Assist Control Valve LF Circuit Failure	ABS Module	CLEAR all DTCs. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE the DTCs. If C1958 is retrieved again along with one or more other ABS valve coil DTCs, go to Pinpoint Test A . If only DTC C1958 is retrieved again, INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
C1963	Stability Control Inhibit Warning	ABS Module	NOTE: REPAIR all other DTCs prior to diagnosing this DTC. This DTC is for information only. CLEAR all DTCs. INSPECT the stability control sensor cluster mounting surface for damage and foreign material. REPAIR or CLEAN as necessary. INSPECT the stability control sensor cluster for correct installation. REFER to Stability Control Sensor Cluster.
C1991	Module Calibration Failure	ABS Module	NOTE: If other DTCs are present, REPAIR them before diagnosing DTC C1991. Go to Pinpoint Test I .
C2769	Longitudinal Acceleration Sensor Electrical Failure	ABS Module	Go to Pinpoint Test F .
C2770	Longitudinal Acceleration Sensor Plausibility Failure	ABS Module	Go to Pinpoint Test F .
U0073	Control Module Communication Bus Off	ABS Module	REFER to MODULE COMMUNICATIONS NETWORK article to diagnose the HS-CAN bus.
U0115	Lost Communication with ECM/PCM - Secondary	ABS Module	REPAIR all other DTCs first. If DTC U0115 is still present, REFER to MODULE COMMUNICATIONS NETWORK article to diagnose the HS-CAN bus.
			REPAIR all other DTCs first. If DTC U0126

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

U0126	Lost Communication with Steering Angle Sensor Module	ABS Module	is still present, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article to diagnose the HS-CAN bus.
U0155	Lost Communication with Instrument Panel Cluster	ABS Module	REPAIR all other DTCs first. If DTC U0155 is still present, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article to diagnose the HS-CAN bus.
U1901	CAN Network #2 Comm Bus Fault - Receive Error	ABS Module	NOTE: If other DTCs are present, REPAIR them before diagnosing DTC U1901. Go to Pinpoint Test J .
U2050	No Application Present	ABS Module	CONFIGURE the ABS module. REFER to <u>MODULE CONFIGURATION</u> article. CLEAR the DTCs. RETRIEVE the DTCs and VERIFY successful module configuration. If DTC U2050 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CLEAR the DTC. REPEAT the self-test.
U2051	One or More Calibration Files Missing or Corrupt	ABS Module	CALIBRATE the ABS module, follow the directions on the scan tool. CLEAR the DTCs. CARRY OUT the ABS module self-test. RETRIEVE and RECORD any DTCs. If DTC U2051 is retrieved again, INSTALL a new ABS module. REFER to <u>Anti-Lock Brake System (ABS) Module</u> . CLEAR the DTC. REPEAT the self-test.

STEERING ANGLE SENSOR MODULE (SASM) DTC CHART

DTC	Description	Source	Action
B1317	Battery Voltage High	Steering Angle Sensor Module (SASM)	NOTE: Battery voltage above 16 volts will cause DTC B1317 to set. Verify battery and charging system condition prior to diagnosing this DTC. Go to Pinpoint Test L .
B1318	Battery Voltage Low	SASM	NOTE: Battery voltage below 10 volts will cause DTC B1318 to set. Verify battery and charging system condition prior to diagnosing this DTC. Go to Pinpoint Test L .

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

B1342	ECU is Faulted	SASM	NOTE: If other DTCs are present, REPAIR them before installing a new component. CLEAR the DTCs. CARRY OUT the SASM self-test. RETRIEVE the DTCs. If DTC B1342 is retrieved again, INSTALL a new SASM. REFER to <u>Steering Angle Sensor Module (SASM)</u>. CLEAR the DTCs. REPEAT the self-test.
U0155	Lost Communication with Instrument Panel Cluster	SASM	REPAIR all other DTCs first. If DTC U0155 is still present, REFER to <u>MODULE COMMUNICATIONS NETWORK</u> article to diagnose the HS-CAN bus.
U2050	No Application Present	SASM	CONFIGURE the SASM. REFER to Programmable Module Installation in <u>MODULE CONFIGURATION</u> article. CLEAR the DTCs. CARRY OUT the SASM self-test. If DTC U2050 is retrieved again, INSTALL a new SASM. REFER to <u>Steering Angle Sensor Module (SASM)</u>. CLEAR the DTC. REPEAT the self-test.

INSTRUMENT CLUSTER DTC CHART

DTC	Description	Source	Action
C1093	Traction Control Disable Switch Circuit Failure	Instrument Cluster	Go to Pinpoint Test K .
-	All Other Instrument Cluster Module DTCs	Instrument Cluster	REFER to <u>INSTRUMENT CLUSTER (IC), MESSAGE CENTER & WARNING CHIMES</u> article.

Symptom Chart

Symptom Chart

Condition	Possible Sources	Action
No communication with the ABS module	- Fuse - Wiring, terminals or connectors - ABS module	REFER to <u>MODULE CONFIGURATION</u> article to diagnose the no communication concern.
The red brake warning indicator is always/never	- Wiring, terminals or connectors - Brake fluid level	RETRIEVE all DTCs and REFER to the <u>ABS MODULE DTC CHART</u> . If no DTCs are present, REFER to Diagnosis and Testing in

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

on	- Instrument cluster - ABS module	<u>INSTRUMENT CLUSTER (IC), MESSAGE CENTER & WARNING CHIMES</u> article.
The yellow ABS warning indicator is always/never on	- Instrument cluster - Brake fluid level - Wiring, terminals or connectors - ABS module	RETRIEVE all DTCs and REFER to the <u>ABS MODULE DTC CHART</u> . If no DTCs are present, REFER to Diagnosis and Testing in <u>INSTRUMENT CLUSTER (IC), MESSAGE CENTER & WARNING CHIMES</u> article.
The traction assist indicator is never/always on	- Instrument cluster - Wiring, terminals or connectors - Traction control switch	RETRIEVE all DTCs and REFER to the <u>ABS MODULE DTC CHART</u> . If no DTCs are present, REFER to Diagnosis and Testing in <u>INSTRUMENT CLUSTER (IC), MESSAGE CENTER & WARNING CHIMES</u> article.
ABS false activation, ABS too sensitive, ABS activates on normal stop	- Front wheel speed sensor tone rings - Chafed wire insulation or pinched wire - Rear wheel speed sensor, excessive variance in air gap between wheel speed sensor and sensor tone ring - Open or shorted wheel speed sensor circuit - Rear wheel speed sensor tone rings - Wheel speed sensor	- INSPECT both front wheel speed sensor tone rings for damaged teeth. INSTALL new as necessary. - INSPECT the wiring harness from the front wheel speed sensor knuckle to the frame and from the rear axle to the frame. REPAIR or INSTALL new as necessary. - INSPECT both rear wheel speed sensor tone rings for damaged teeth. INSTALL new as necessary. - Go to Pinpoint Test C . - INSPECT the rear wheel speed sensor tone rings. INSTALL new as necessary. - Go to Pinpoint Test D .
The traction control system cannot be disabled	- Instrument cluster - Circuitry - Traction control switch	Go to Pinpoint Test K .

Pinpoint Tests**Pinpoint Test A: DTCs B1317 and B1318 - Battery Voltage High/Low**

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The operating voltage required to operate the ABS module, hydraulic valves and pump motor is between 10 and 16 volts.

Fused ignition voltage is supplied to the ABS module from battery junction box (BJB) fuse 50 (10A) along circuit CBB50 (YE/OG). Fused battery voltage is supplied to the ABS module from BJB fuse 57 (40A) along circuit SBB57 (RD) and BJB fuse 15 (40A) along circuit SBB15 (WH/RD). Ground is provided for the ABS module along circuit GD121 (BK/YE).

- DTC B1317 Battery Voltage High - If the voltage supplied to the ABS module exceeds 16 volts during the ABS module self-test or normal operation, with vehicle speed greater than 5 km/h (3 mph), DTC B1317 will be set.
- DTC B1318 Battery Voltage Low - If the voltage supplied to the ABS module falls below 10 volts during the ABS module self-test or normal operation, with vehicle speed greater than 5 km/h (3 mph), DTC B1318 will be set.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Charging system
- ABS module

PINPOINT TEST A: DTCs B1317 AND B1318 - BATTERY VOLTAGE HIGH/LOW

NOTE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

A1 CHECK FOR RETURNING DTCs

- Connect the diagnostic tool.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Clear ABS Module DTCs
- Enter the following diagnostic mode on the diagnostic tool: Self-Test ABS Module
- Carry out the ABS module self-test. Retrieve and record all ABS module DTCs.
- **Are DTCs B1317 or 1318 present?**

YES : Go to A2.

NO : REFER to **CHARGING SYSTEM - GENERAL INFORMATION** article to continue

diagnosis of the charging system.

A2 CHECK THE BATTERY VOLTAGE

- Measure the voltage between the positive and negative battery terminals with key ON engine OFF (KOEO) and with the engine running.
- **Is the battery voltage between 10 and 13 volts with KOEO and between 13 and 17 volts with the engine running?**

YES : Go to A3.

NO : REFER to **CHARGING SYSTEM - GENERAL INFORMATION** article to continue diagnosis of the charging system.

A3 CHECK THE ABS HARNESS AND MODULE FOR WATER INTRUSION

NOTE: The battery and battery tray need to be removed to access the ABS module connector.

- Key in OFF position.
- Disconnect: ABS Module C135
- Inspect the ABS module connector and module for water intrusion.
- **Is there any evidence of water entry?**

YES : CLEAN the connector and dry it with low pressure compressed air. INSPECT for evidence of water entry and REPAIR as necessary. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

NO : Go to A4.

A4 CHECK THE ABS MODULE POWER CIRCUITS FOR AN OPEN

- Disconnect: BJB Fuse 15 (40A)
- Disconnect: BJB Fuse 50 (10A)
- Disconnect: BJB Fuse 57 (40A)

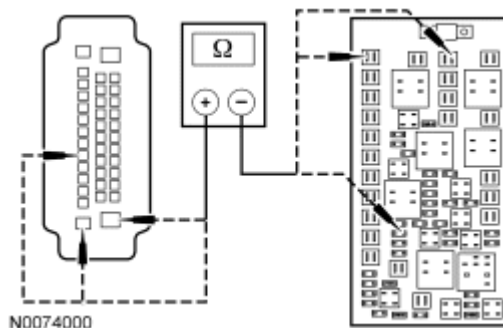


Fig. 1: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between:
 - ABS module C135-1, circuit SBB15 (WH/RD), harness side and BJB fuse 15 (40A), circuit SBB15 (WH/RD), output side.

- ABS module C135-25, circuit SBB57 (RD), harness side and BJB fuse 57 (40A), circuit SBB57 (RD), output side.
- ABS module C135-30, circuit CBB50 (YE/OG), harness side and BJB fuse 50 (10A), circuit CBB50 (YE/OG), output side.
- **Are the resistances less than 5 ohms?**
YES : Go to A5.
NO : VERIFY BJB fuses 50 (10A), 57 (40A) and 15 (40A) are OK. If OK, REPAIR the affected circuit(s). CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

A5 CHECK THE ABS MODULE GROUND CIRCUITS FOR AN OPEN

- Key in OFF position.

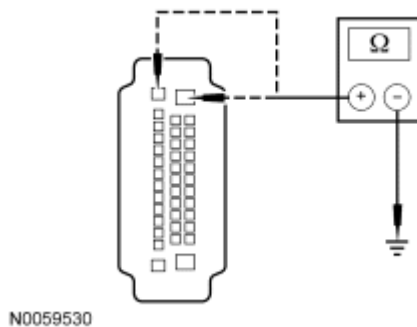


Fig. 2: Measuring Resistance Between Ground & ABS Module C135-13 & C135-38, Circuit GD121 (BK/YE)
 Courtesy of FORD MOTOR CO.

- Measure the resistance between ground and:
 - ABS module C135-13, circuit GD121 (BK/YE), harness side.
 - ABS module C135-38, circuit GD121 (BK/YE), harness side.
- **Are the resistances less than 5 ohms?**
YES : Go to A6.
NO : REPAIR circuit GD121 (BK/YE). CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

A6 CHECK FOR CORRECT ABS MODULE OPERATION

- Check ABS module C135 for:
 - corrosion.
 - pushed-out pins.
 - spread terminals.
- Connect: ABS C135
- Make sure the ABS connector seats correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

YES : INSTALL a new ABS module. REFER to **Anti-Lock Brake System (ABS) Module**. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

Pinpoint Test B: DTCs C1095 and C1096 - ABS Hydraulic Pump Motor Circuit Failure/Open

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

Fused battery voltage for hydraulic pump motor operation is supplied to the ABS module from battery junction box (BJB) fuse 15 (40A) along circuit SBB15 (WH/RD). Ground is provided for the ABS module along circuit GD121 (BK/YE).

When the ignition key is turned to the ON position, the ABS module commands the HCU pump on for 100 ms (± 6 ms) and is then commanded off. After 6 ms, the ABS module will read the voltage that is being generated by the HCU pump. If the voltage indicates the motor is spinning at less than 500 RPM, there may be a locked motor. If this condition is detected 4 times, DTC C1095 is set. The pump motor is checked for an open circuit 2 seconds after the most recent successful pump motor off command. If the pump motor feedback remains greater than 0.75 volt for more than 50 ms (± 6 ms) after these conditions have been met, DTC C1096 is set.

- DTC C1095 ABS Hydraulic Pump Motor Circuit Failure - If a short to voltage or ground is detected, if a locked up pump motor is detected or if there is an internal failure of the ABS module, DTC C1095 will be set.
- DTC C1096 ABS Hydraulic Pump Motor Circuit Open - If a short to voltage, a short to ground or an open is detected or if there is an internal failure of the ABS module, DTC C1095 will be set.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- HCU
- ABS module

PINPOINT TEST B: DTCs C1095 AND C1096 - ABS HYDRAULIC PUMP MOTOR CIRCUIT FAILURE/OPEN

NOTE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

B1 CHECK THE ABS PUMP MOTOR

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

- Key in ON position.

- **Is the ABS pump motor running all the time?**

YES : INSTALL a new ABS module and hydraulic control unit (HCU). REFER to **Anti-Lock Brake System (ABS) Module** and **Hydraulic Control Unit (HCU)**. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

NO : Go to B2.

B2 CHECK ABS MODULE PUMP MOTOR OUTPUT COMMAND (PUMP_MTR)

- Enter the following diagnostic mode on the diagnostic tool: DataLogger ABS Module
- Toggle the PUMP_MTR output command ON.
- **Does the ABS pump motor run for approximately 2 seconds?**

YES : TOGGLE the output command OFF. CLEAR the DTCs. CHECK the yellow ABS warning indicator while driving the vehicle (brakes must not be applied) above 32 km/h (20 mph).

If the yellow ABS warning indicator does not illuminate, the system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

If the yellow ABS warning indicator illuminates, RETRIEVE the DTCs.

If DTC C1096 is retrieved, go to B6.

If DTC C1095 is retrieved, INSTALL a new HCU. REFER to **Hydraulic Control Unit (HCU)**. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

NO : TOGGLE the output command OFF. Go to B3.

B3 CHECK THE ABS HARNESS AND MODULE FOR WATER INTRUSION

NOTE: **The battery and battery tray need to be removed to access the ABS module connector.**

- Key in OFF position.
- Disconnect: ABS Module C135
- Inspect the ABS module connector and module for water intrusion.
- **Is there any evidence of water entry?**

YES : CLEAN the connector and dry it with low pressure compressed air. INSPECT for evidence of water entry and REPAIR as necessary. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

NO : Go to B4.

B4 CHECK CIRCUIT SBB15 (WH/RD) FOR AN OPEN

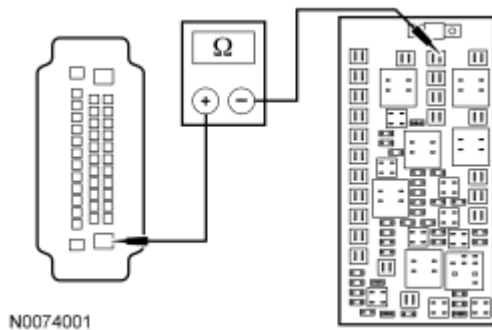


Fig. 3: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between ABS module C135-1, circuit SBB15 (WH/RD), harness side and BJB fuse 15 (40A), circuit SBB15 (WH/RD), output side.
- **Is the resistance less than 5 ohms?**
YES : Go to B5.
NO : VERIFY BJB fuse 15 (40A) is OK. If OK, REPAIR circuit SBB15 (WH/RD). CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

B5 CHECK CIRCUIT GD121 (BK/YE) FOR AN OPEN

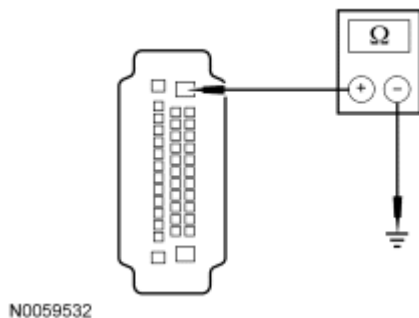


Fig. 4: Measuring Resistance Between ABS Module C135-13, Circuit GD121 (BK/YE), Harness Side & Ground
Courtesy of FORD MOTOR CO.

- Measure the resistance between ABS module C135-13, circuit GD121 (BK/YE), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : INSTALL a new HCU. REFER to **Hydraulic Control Unit (HCU)**. CLEAR the DTCs. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
NO : REPAIR circuit GD121 (BK/YE). CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

B6 CHECK FOR CORRECT ABS MODULE OPERATION

- Check ABS module C135 for:

- corrosion.
- pushed-out pins.
- spread terminals.
- Connect: ABS C135
- Make sure the ABS connector seats correctly.
- Clear the DTCs.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ABS module. REFER to **Anti-Lock Brake System (ABS) Module**. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

Pinpoint Test C: DTCs C1145, C1155, C1165 and C1175 - Wheel Speed Sensor Input Circuit Failure

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The wheel speed sensors generate a square wave signal that is sent to the ABS module. The wheel speed sensor circuitry connects to the ABS module through 2 wires and a connector at each wheel speed sensor. When the ignition is turned to the RUN position, the ABS module carries out a self-test to determine if the sensors are functional. Voltage is supplied to the active wheel speed sensors from the ABS module. The wheel speed sensor ground is supplied by the ABS module.

Wheel speed sensor DTCs will automatically clear when a successful test drive is carried out. A successful test drive should include speeds above 32 km/h (20 mph) and at least one ABS stop.

DTC Description	Fault Trigger Condition
• DTC C1145 RF Wheel Speed Sensor Input Circuit Failure • DTC C1155 LF Wheel Speed Sensor Input Circuit Failure • DTC C1165 RR Wheel Speed Sensor Input Circuit Failure • DTC C1175 LR Wheel Speed Sensor Input Circuit Failure	If during the ABS module on-demand self test or during normal operation, if the ABS module detects an open, short to ground or voltage or a defective wheel speed sensor input circuit, the appropriate DTC will be set. These DTCs are usually set by a damaged sensor or a circuit fault.

This pinpoint test is intended to diagnose the following:

- Connector water intrusion
- Wiring, terminals or connectors
- Wheel speed sensor tone ring
- Wheel speed sensors
- ABS module

PINPOINT TEST C: DTCs C1145, C1155, C1165 AND C1175 - WHEEL SPEED SENSOR INPUT CIRCUIT FAILURE

NOTE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

C1 CHECK THE ABS HARNESS AND MODULE FOR WATER INTRUSION

NOTE: The battery and battery tray need to be removed to access the ABS module connector.

- Key in OFF position.
- Disconnect: ABS Module C135
- Inspect the ABS module connector and module for water intrusion.
- **Is there any evidence of water entry?**

YES : CLEAN the connector and dry it with low pressure compressed air. INSPECT for evidence of water entry and REPAIR as necessary. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

NO : If the active wheel speed sensor tool is available, go to C2. If the active wheel speed sensor tool is not available, go to C4.

C2 CHECK THE ABS MODULE OUTPUT USING THE SPECIAL TOOL

NOTE: The front wheel speed sensor connectors are located behind the fender splash shields.

- Connect: ABS Module C135
- Disconnect: Suspect Wheel Speed Sensor
- Connect the special tool to the wheel speed sensor connectors.
- Key in ON position.
- Select the correct system polarity on the special tool and turn the special tool power switch to the ON position.
- **Is the module output LED illuminated?**

YES : Go to C3.

NO : Go to C6.

C3 CHECK THE WHEEL SPEED SENSOR OUTPUT WITH THE SPECIAL TOOL

- Raise the suspect wheel until it can spin freely. Refer to **JACKING AND LIFTING** article.
- While monitoring the special tool, slowly spin the suspect wheel.
- **Is the current level LED illuminated green and do the sensor output LEDs illuminate and flash?**

YES : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

NO : If the current level LED is illuminated green and the sensor output LEDs do not illuminate or if the current level LED is illuminated red, INSTALL a new wheel speed sensor. REFER to **Wheel Speed Sensor - Front** or **Wheel Speed Sensor - Rear**. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

If the current level LED is illuminated green and the sensor output LEDs illuminate green but do not flash, INSPECT the wheel speed sensor tone ring and INSTALL new if necessary. If the tone ring is OK, INSTALL a new wheel speed sensor. REFER to **Wheel Speed Sensor - Front** or **Wheel Speed Sensor - Rear**. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

C4 CHECK THE WHEEL SPEED SENSOR CIRCUITS FOR A SHORT TO VOLTAGE

NOTE: Both circuits must be checked for each DTC.

NOTE: The battery must be connected to carry out this test.

NOTE: The front wheel speed sensor connectors are located behind the fender splash shields.

- Disconnect: Suspect Wheel Speed Sensor
- Key in ON position.
- **For DTC C1145** , measure the voltage between ground and:
 - ABS module C135-3, circuit VCA05 (GY/VT), harness side.
 - ABS module C135-4, circuit RCA19 (VT), harness side.
- **For DTC C1155** , measure the voltage between ground and:
 - ABS module C135-27, circuit VCA03 (VT/WH), harness side.
 - ABS module C135-28, circuit RCA17 (YE), harness side.
- **For DTC C1165** , measure the voltage between ground and:
 - ABS module C135-10, circuit RCA20 (BN), harness side.
 - ABS module C135-11, circuit VCA06 (WH/OG), harness side.
- **For DTC C1175** , measure the voltage between ground and:

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

- ABS module C135-35, circuit RCA18 (BN/GN), harness side.
- ABS module C135-36, circuit VCA04 (BU/OG), harness side.
- **Is any voltage present?**
YES : REPAIR the affected circuit(s). TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.
NO : Go to C5.

C5 CHECK THE WHEEL SPEED SENSOR CIRCUITS FOR A SHORT TO GROUND

NOTE: Both circuits must be checked for each DTC.

- Key in OFF position.
- **For DTC C1145** , measure the resistance between ground and:
 - ABS module C135-3, circuit VCA05 (GY/VT), harness side.
 - ABS module C135-4, circuit RCA19 (VT), harness side.
- **For DTC C1155** , measure the resistance between ground and:
 - ABS module C135-27, circuit VCA03 (VT/WH), harness side.
 - ABS module C135-28, circuit RCA17 (YE), harness side.
- **For DTC C1165** , measure the resistance between ground and:
 - ABS module C135-10, circuit RCA20 (BN), harness side.
 - ABS module C135-11, circuit VCA06 (WH/OG), harness side.
- **For DTC C1175** , measure the resistance between ground and:
 - ABS module C135-35, circuit RCA18 (BN/GN), harness side.
 - ABS module C135-36, circuit VCA04 (BU/OG), harness side.
- **Are the resistances greater than 10,000 ohms?**
YES : Go to C6.
NO : REPAIR the affected circuit(s). TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

C6 CHECK THE WHEEL SPEED SENSOR CIRCUITS FOR AN OPEN

NOTE: Both circuits must be checked for each DTC.

- Disconnect: ABS Module C135
- Disconnect: Suspect Wheel Speed Sensor
- **For DTC C1145** , measure the resistance between:
 - ABS module C135-3, circuit VCA05 (GY/VT), harness side and RF wheel speed sensor C160-1, circuit VCA05 (GY/VT), harness side.
 - ABS module C135-4, circuit RCA19 (VT), harness side and RF wheel speed sensor C160-2, circuit RCA19 (VT), harness side.
- **For DTC C1155** , measure the resistance between:

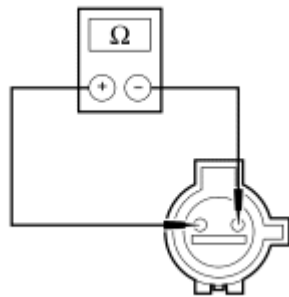
- ABS module C135-27, circuit VCA03 (VT/WH), harness side LF wheel speed sensor C150-2, circuit VCA03 (VT/WH), harness side.
- ABS module C135-28, circuit RCA17 (YE), harness side LF wheel speed sensor C150-1, circuit RCA17 (YE), harness side.
- **For DTC C1165** , measure the resistance between:
 - ABS module C135-10, circuit RCA20 (BN), harness side and RR wheel speed sensor C426-2, circuit RCA20 (BN), harness side.
 - ABS module C135-11, circuit VCA06 (WH/OG), harness side and RR wheel speed sensor C426-1, circuit VCA06 (WH/OG), harness side.
- **For DTC C1175** , measure the resistance between:
 - ABS module C135-35, circuit RCA18 (BN/GN), harness side and LR wheel speed sensor C440-1, circuit RCA18 (BN/GN), harness side.
 - ABS module C135-36, circuit VCA04 (BU/OG), harness side and LR wheel speed sensor C440-2, circuit VCA04 (BU/OG), harness side.

- **Are the resistances less than 5 ohms?**

YES : Go to C7.

NO : REPAIR the affected circuit(s). TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

C7 CHECK FOR WHEEL SPEED SENSOR CIRCUITS SHORTED TOGETHER



A0057968

Fig. 5: Checking Circuits For Short
Courtesy of FORD MOTOR CO.

- Measure the resistance between the suspect wheel speed sensor pins, harness side, with the meter in the 10 megaohm range.
- **Is the resistance greater than 10,000 ohms?**

YES : Go to C8.

NO : REPAIR the affected circuit(s). TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

C8 CHECK THE ABS MODULE OUTPUT

- Connect: ABS Module C135
- Key in ON position.

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

- Measure the voltage between the suspect wheel speed sensor pins, harness side.
- **Is the voltage greater than 10 volts?**

YES : INSTALL a new wheel speed sensor. REFER to Wheel Speed Sensor - Front or Wheel Speed Sensor - Rear. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

NO : Go to C9.

C9 CHECK FOR CORRECT ABS MODULE OPERATION

- Key in OFF position.
- Disconnect: ABS C135
- Check the connector for:
 - corrosion.
 - pushed-out pins.
 - spread terminals.
- Connect: ABS C135
- Make sure the ABS connector seats correctly.
- Test drive the vehicle above 32 km/h (20 mph), carry out at least one ABS stop and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ABS module. REFER to Anti-Lock Brake System (ABS) Module. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

Pinpoint Test D: DTCs C1222, C1233, C1234, C1235 and C1236 - Wheel Speed Sensor Input Signal Missing/Mismatch

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The wheel speed sensors generate a square wave signal that is sent to the ABS module. The wheel speed sensor circuitry connects to the ABS module through 2 wires and a connector at each wheel speed sensor. When the ignition is turned to the RUN position, the ABS module carries out a self-test to determine if the sensors are functional. Voltage is supplied to the active wheel speed sensors from the ABS module. The wheel speed sensor ground is supplied by the ABS module.

Wheel speed sensor DTCs will automatically clear when a successful test drive is carried out. A successful test drive should include speeds above 32 km/h (20 mph) and at least one ABS stop.

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

DTC Description	Fault Trigger Condition
• DTC C1222 Wheel Speed Mismatch	When the vehicle speed exceeds 20 km/h (12 mph), if the ABS module detects a difference between wheel speed sensor signals, DTC C1222 will be set. DTC C1222 can also be set by damaged tone rings, mismatched wheel and/or tire sizes or driving the vehicle on one or more deflated tires
• DTC C1233 LF Wheel Speed Sensor Signal Fault • DTC C1234 RF Wheel Speed Sensor Signal Fault • DTC C1235 RR Wheel Speed Sensor Signal Fault • DTC C1236 LR Wheel Speed Sensor Signal Fault	When the vehicle speed exceeds 20 km/h (12 mph), if the ABS module does not receive a signal from the wheel speed sensor, the appropriate DTC will be set. These DTCs can also be set by an excessive wheel speed sensor-to-tone ring air gap or a damaged tone ring

This pinpoint test is intended to diagnose the following:

- Mismatched tire sizes
- Wheel speed sensor tone ring
- Wheel speed sensor
- ABS module

PINPOINT TEST D: DTCs C1222, C1233, C1234, C1235 AND C1236 - WHEEL SPEED SENSOR INPUT SIGNAL MISSING/MISMATCH

NOTE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

D1 RETRIEVE ABS MODULE DTCs

- Connect the diagnostic tool.
- Key in ON position.
- Retrieve and record all ABS module DTCs.
- **Are DTCs C1145, C1155, C1165 or C1175 present?**
YES : Go to **Pinpoint Test C** .
NO : Go to D2.

D2 CHECK THE ABS HARNESS AND MODULE FOR WATER INTRUSION

NOTE: The battery and battery tray need to be removed to access the ABS module connector.

- Key in OFF position.

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

- Disconnect: ABS Module C135
- Inspect the ABS module connector and module for water intrusion.
- **Is there any evidence of water entry?**
YES : CLEAN the connector and dry it out with low pressure compressed air. INSPECT for evidence of water entry and repair as necessary. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.
NO : Use the DTCs recorded in the previous step.

For DTC C1222 Go to D7. For DTCs C1233, C1234, C1235 and C1236, if the active wheel speed sensor tool is available, go to D3. If the active wheel speed sensor tool is not available, go to D6. For all other DTCs, Go to the **ABS MODULE DTC CHART** .

D3 CHECK THE ABS MODULE OUTPUT USING THE SPECIAL TOOL

NOTE: **The front wheel speed sensor connectors are located behind the fender splash shields.**

- Connect: ABS Module C135
- Disconnect: Suspect Wheel Speed Sensor
- Connect the special tool to the wheel speed sensor connectors.
- Key in ON position.
- Select the correct system polarity on the special tool and turn the special tool power switch to the ON position.
- **Is the module output LED illuminated?**
YES : Go to D5.
NO : Go to D4.

D4 CHECK THE WHEEL SPEED SENSOR CIRCUITS FOR AN OPEN

NOTE: **Both circuits must be checked for each DTC.**

- Key in OFF position.
- Disconnect: ABS Module C135
- **For DTC C1233** : measure the resistance between:
 - ABS module C135-27, circuit VCA03 (VT/WH), harness side and LF wheel speed sensor C150-2, circuit VCA03 (VT/WH), harness side.
 - ABS module C135-28, circuit RCA17 (YE), harness side and LF wheel speed sensor C150-1, circuit RCA17 (YE), harness side.
- **For DTC C1234** : measure the resistance between:
 - ABS module C135-3, circuit VCA05 (GY/VT), harness side and RF wheel speed sensor C160-1, circuit VCA05 (GY/VT), harness side.
 - ABS module C135-4, circuit RCA19 (VT), harness side and RF wheel speed sensor C160-2,

circuit RCA19 (VT), harness side.

- **For DTC C1235** : measure the resistance between:
 - ABS module C135-10, circuit RCA20 (BN), harness side and RR wheel speed sensor C426-2, circuit RCA20 (BN), harness side.
 - ABS module C135-11, circuit VCA06 (WH/OG), harness side and RR wheel speed sensor C426-1, circuit VCA06 (WH/OG), harness side.
- **For DTC C1236** : measure the resistance between:
 - ABS module C135-35, circuit RCA18 (BN/GN), harness side and LR wheel speed sensor C440-1, circuit RCA18 (BN/GN), harness side.
 - ABS module C135-36, circuit VCA04 (BU/OG), harness side and LR wheel speed sensor C440-2, circuit VCA04 (BU/OG), harness side.

- **Are the resistances less than 5 ohms?**

YES : Go to D8.

NO : REPAIR the affected circuit(s). TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

D5 CHECK THE WHEEL SPEED SENSOR OUTPUT WITH THE SPECIAL TOOL

- Raise the suspect wheel until it can spin freely. Refer to **JACKING AND LIFTING** article.
- While monitoring the special tool, slowly spin the suspect wheel.
- **Is the current level LED illuminated green and do the sensor output LEDs illuminate and flash?**

YES : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

NO : If the current level LED is illuminated green and the sensor output LEDs do not illuminate or if the current level LED is illuminated red, INSTALL a new wheel speed sensor. REFER to **Wheel Speed Sensor - Front** or **Wheel Speed Sensor - Rear**. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

If the current level LED is illuminated green and the sensor output LEDs illuminate green but do not flash, INSPECT the wheel speed sensor tone ring and INSTALL new if necessary. If the tone ring is OK, INSTALL a new wheel speed sensor. REFER to **Wheel Speed Sensor - Front** or **Wheel Speed Sensor - Rear**. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

D6 CHECK THE WHEEL SPEED SENSOR OUTPUT

NOTE: **The front wheel speed sensor connectors are located behind the fender splash shields.**

NOTE: **The battery must be connected to carry out this test.**

- Connect: ABS Module C135

- Disconnect: Suspect Wheel Speed Sensor

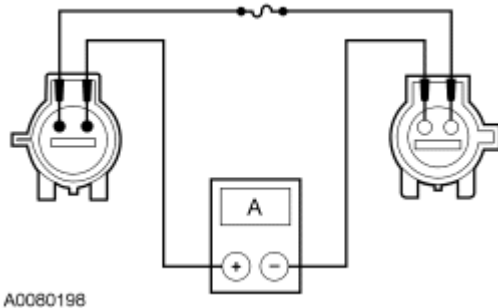


Fig. 6: Checking Wheel Speed Sensor Output
Courtesy of FORD MOTOR CO.

- **For DTC C1233 :**
 - connect a fused (5A) jumper wire between LF wheel speed sensor C150-2, circuit VCA03 (VT/WH), harness side and LF wheel speed sensor C150 pin-1, component side.
 - connect the automotive meter between LF wheel speed sensor C150-1, circuit RCA17 (YE), harness side and LF wheel speed sensor C150 pin-2, component side.
- **For DTC C1234 :**
 - connect a fused (5A) jumper wire between RF wheel speed sensor C160-1, circuit VCA05 (GN/VT), harness side and RF wheel speed sensor C160 pin-1, component side.
 - connect the automotive meter between RF wheel speed sensor C160-2, circuit RCA19 (VT), harness side and RF wheel speed sensor C160 pin-2, component side.
- **For DTC C1235 :**
 - connect a fused (5A) jumper wire between RR wheel speed sensor C426-1, circuit VCA06 (WH/OG), harness side and RR wheel speed sensor C420 pin-1, component side.
 - connect the automotive meter between RR wheel speed sensor C426-2, circuit RCA20 (BN), harness side and RR wheel speed sensor C420 pin-2, component side.
- **For DTC C1236 :**
 - connect a fused (5A) jumper wire between LF wheel speed sensor C440-2, circuit VCA04 (BU/OG), harness side and LF wheel speed sensor C440 pin-1, component side.
 - connect the automotive meter between LF wheel speed sensor C440-1, circuit RCA18 (BN/GN), harness side and LF wheel speed sensor C440 pin-2, component side.
- Key in ON position.
- Measure the current while slightly rotating the wheel, using incremental turns.
- **Does the wheel speed sensor output switch between the low state (4 to 8 mA) and the high state (12 to 16 mA)?**
YES : Go to D7.
NO : INSTALL a new wheel speed sensor. REFER to Wheel Speed Sensor - Front or Wheel Speed Sensor - Rear. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

D7 CHECK FOR CORRECT TIRE SIZE AND AIR PRESSURE

- Make sure all tire sizes are correct and matching.
- Make sure all tire pressures are correct.
- Clear the DTCs, drive the vehicle and carry out the self-test.
- **Is DTC C1222 still present?**

YES : Go to D8.

NO : The system is operating correctly at this time. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

D8 CHECK THE ABS MODULE ELECTRICAL CONNECTOR

- Key in OFF position.
- Disconnect: ABS C135
- Check the connector for:
 - corrosion.
 - pushed-out pins.
 - spread terminals.
- Connect: ABS C135
- Make sure the ABS connector seats correctly.
- Test drive the vehicle above 32 km/h (20 mph), carry out at least one ABS stop and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new ABS module. REFER to **Anti-Lock Brake System (ABS) Module**. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop.

Pinpoint Test E: DTC C1278 - Steering Wheel Angle 1 and 2 Circuit Faulted

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The steering angle sensor module (SASM) (also known as the steering angle sensor module) monitors the steering wheel position when the slotted-wheel in the sensor passes through the optical field of the sensor.

Fused ignition voltage is supplied to the SASM from smart junction box (SJB) fuse 34 (5A) along circuit CBP34 (VT/BN). Ground for the sensor is provided along circuit GD115 (BK/GY). The steering wheel angle (in degrees) is sent to the ABS module along the high-speed controller area network (HS-CAN) bus, circuits VDB04 (WH/BU) and VDB05 (WH).

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

- DTC C1278 Steering Wheel Angle 1 and 2 Circuit Faulted - If during normal operation or the ABS module on-demand self test, the ABS module does not receive the steering wheel rotation sensor input or if the input received does not agree with the input from other sensors, DTC C1278 will be set. Also, a loss of sensor power or ground will cause DTC C1278 to set.

This pinpoint test is intended to diagnose the following:

- Fuse(s)
- Wiring, terminals or connectors
- SASM

PINPOINT TEST E: DTCs C1277 AND C1278 - STEERING WHEEL ANGLE 1 AND 2 CIRCUIT FAILURE/FAULTED

NOTE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

NOTE: Overtightening of the steering angle sensor module (SASM) fasteners may deform or damage the sensor housing, possibly resulting in faulty operation.

NOTE: Make sure that the SASM is mounted securely to the column with 2 fasteners. A loose sensor or missing fastener may result in incorrect sensor operation. Make sure that the sensor connector is mated securely to the harness connector. Before proceeding with electrical testing, make sure that the vehicle battery is fully charged.

E1 CARRY OUT THE CALIBRATION PROCEDURE

- Connect the diagnostic tool.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: ABS Module Calibration
- Carry out the ABS module calibration procedure following the scan tool directions.
- Test drive the vehicle.
- Retrieve and record any ABS module DTCs.
- **Are any DTCs retrieved or does the calibration procedure indicate a fault?**

YES : If DTC C1277 or C1278 is retrieved or the calibration procedure indicates a fault, go to E2.

If any other DTCs are retrieved, go to the **ABS MODULE DTC CHART** .

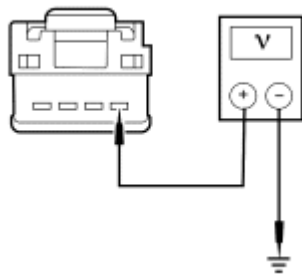
NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

E2 MONITOR THE SASM PID (SWA POS)

- Enter the following diagnostic mode on the diagnostic tool: DataLogger ABS Module
- Monitor the SWA POS PID while rotating the steering wheel clockwise and counterclockwise very slowly.
- **Does the PID move in intervals of 1 degree?**
YES : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.
NO : Go to E3.

E3 CHECK THE VOLTAGE TO THE SASM

- Key in OFF position.
- Disconnect: SASM C226
- Key in ON position.



N0059533

Fig. 7: Measuring Voltage Between Steering Angle Sensor C222-1, CBP34 (VT/BN), Harness Side & Ground

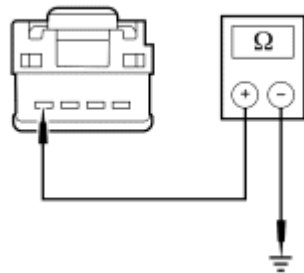
Courtesy of FORD MOTOR CO.

- Measure the voltage between SASM C226-1, circuit CBP34 (VT/BN), harness side and ground.
- **Is the voltage greater than 10 volts?**
YES : Go to E4.
NO : VERIFY SJB fuse 34 (5A) are OK. If OK, REPAIR circuit CBP34 (VT/BN).

CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

E4 CHECK CIRCUIT GD115 (BK/GY) FOR AN OPEN

- Key in OFF position.



N0059535

Fig. 8: Measuring Resistance Between Steering Angle Sensor C222-4, GD115 (BK/GY), Harness Side & Ground
Courtesy of FORD MOTOR CO.

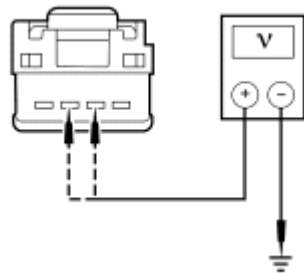
- Measure the resistance between SASM C226-4, circuit GD115 (BK/GY), harness side and ground.
- **Is the resistance less than 5 ohms?**

YES : Go to E5.

NO : REPAIR circuit GD115 (BK/GY). CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

E5 CHECK CIRCUITS VDB04 (WH/BU) AND VDB05 (WH) FOR A SHORT TO VOLTAGE

- Key in ON position.



N0059534

Fig. 9: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between ground and:
 - SASM C226-3, circuit VDB04 (WH/BU), harness side.
 - SASM C226-2, circuit VDB05 (WH), harness side.

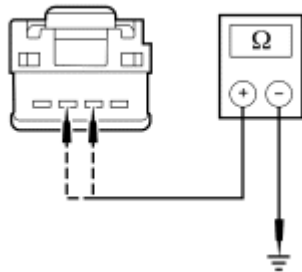
- **Are the voltages greater than 6 volts?**

YES : REPAIR the affected circuit(s). CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

NO : Go to E6.

E6 CHECK CIRCUITS VDB04 (WH/BU) AND VDB05 (WH) FOR A SHORT TO GROUND

- Key in OFF position.



N0059536

Fig. 10: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between ground and:
 - SASM C226-3, circuit VDB04 (WH/BU), harness side.
 - SASM C226-2, circuit VDB05 (WH), harness side.

- **Are the resistances greater than 10,000 ohms?**

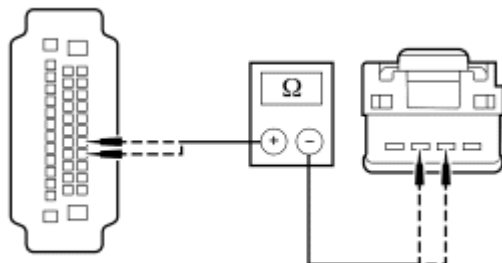
YES : Go to E7.

NO : REPAIR the affected circuit(s). CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

E7 CHECK CIRCUITS VDB04 (WH/BU) AND VDB05 (WH) FOR AN OPEN

NOTE: The battery and battery tray need to be removed to access the ABS module connector.

- Disconnect: ABS Module C135



N0059537

Fig. 11: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between:

- ABS module C135-5, circuit VDB04 (WH/BU), harness side and SASM C226-3, circuit VDB04 (WH/BU), harness side.
- ABS module C135-6, circuit VDB05 (WH), harness side and SASM C226-2, circuit VDB05 (WH), harness side.
- **Are the resistances less than 5 ohms?**

YES : Go to E8.

NO : REPAIR the affected circuit(s). CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

E8 CHECK CIRCUITS VDB04 (WH/BU) AND VDB05 (WH) FOR A SHORT TOGETHER

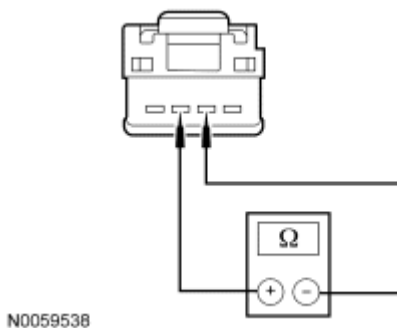


Fig. 12: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between SASM C226-3, circuit VDB04 (WH/BU), harness side and SASM C226-2, circuit VDB05 (WH), harness side.
- **Is the resistance greater than 10,000 ohms?**

YES : INSTALL a new SASM. REFER to **Steering Angle Sensor Module (SASM)**.

CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

NO : REPAIR the affected circuit(s). CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

Pinpoint Test F: DTC C1279, C1280, C1281, C1282, C1516, C1517, C2769, C2770 - Stability Control Sensor Cluster Fault

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The accelerometer (lateral and longitudinal) and the yaw rate sensor are contained in the stability control sensor cluster. The sensor cluster measures the vehicle acceleration and yaw rate, then sends the signal along a secondary high-speed controller area network (HS-CAN2) bus to the ABS module, circuits VCA23 (BU/WH)

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

and VCA24 (GN/OG). Fused ignition voltage is supplied to the sensor cluster by smart junction box (SJB) fuse 35 (10A) along circuit CBP35 (YE/GY). Ground for the sensor cluster is provided along circuit GD151 (BK/GN).

DTC Description	Fault Trigger Conditions
• DTC C1279 Yaw Rate Sensor Circuit Failure • DTC C1281 Lateral Accelerometer Circuit Failure • DTC C1517 Roll Rate Sensor Circuit Failure • DTC C2769 Longitudinal Acceleration Sensor Electrical Failure	If, during normal operation or the ABS module on-demand self test the ABS module detects an open circuit, a short to voltage or a short to ground on one or both of the HS-CAN2 bus circuits, the appropriate DTC will be set. Also, a failure of the sensor cluster will cause the appropriate DTC to set.
• DTC C1280 Yaw Rate Sensor Signal Fault • DTC C1282 Lateral Accelerometer Signal Fault • DTC C1516 Roll Rate Sensor Signal Fault • DTC C2770 Longitudinal Acceleration Sensor Plausibility Failure	If, during normal operation the ABS module does not receive the sensor cluster input, or if the input received does not agree with the input from other sensors, the appropriate DTC will be set.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Stability control sensor cluster

PINPOINT TEST F: DTC C1279, C1280, C1281, C1282, C1516, C1517, C2769, C2770 - STABILITY CONTROL SENSOR CLUSTER FAULT

NOTE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

NOTE: The yaw rate sensor, roll-rate sensor, lateral accelerometer and longitudinal accelerometer are integral to the stability control sensor cluster.

NOTE: Make sure the vehicle is sitting on level ground, at a complete standstill and is not on a hoist before calibrating the stability control sensor cluster.

NOTE: Make sure the vehicle is sitting on level ground, at a complete standstill and is not on a hoist before carrying out this pinpoint test. Any vehicle movement will result in false values for this test.

F1 CARRY OUT THE CALIBRATION PROCEDURE

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: ABS Module Calibration
- Carry out the ABS module calibration procedure following the scan tool directions.
- Test drive the vehicle.
- Retrieve and record any ABS module DTCs.
- **Are any DTCs retrieved or does the calibration procedure indicate a fault?**

YES : Go to F2.

NO : The system is operating correctly. The condition may have been caused by an incomplete or inaccurate stability control sensor cluster calibration.

F2 CHECK THE SENSOR CLUSTER MOUNTING AND MOUNTING SURFACE

NOTE: The sensor cluster is located on the RH side of the vehicle, under the rear seat.

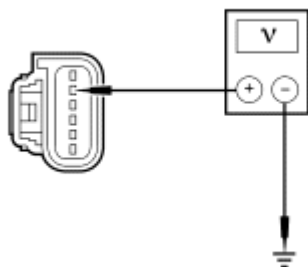
- Key in OFF position.
- Inspect the sensor cluster mounting and make sure that the nuts are fully seated and tightened correctly. Inspect the mounting surface for damage, corrosion or dirt.
- **Were the sensor cluster nuts tightened incorrectly or a significant amount of corrosion or dirt found or the mounting surface damaged?**

YES : CLEAN and TIGHTEN the nuts to correct specification. CLEAN or REPAIR the mounting surface as necessary. CLEAR the DTCs. RECALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. REPEAT the self-test.

NO : Go to F3.

F3 CHECK THE VOLTAGE TO THE SENSOR CLUSTER

- Disconnect: Stability Control Sensor Cluster C3037
- Key in ON position.



N0059539

Fig. 13: Measuring Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between the sensor cluster C3037-2, circuit CBP35 (YE/GY), harness side and ground.

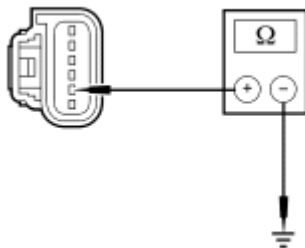
- **Is the voltage greater than 10 volts?**

YES : Go to F4.

NO : VERIFY SJB fuse 35 (10A) is OK. If OK, REPAIR circuit CBP35 (YE/GY). CLEAR the DTCs. RECALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. REPEAT the self-test.

F4 CHECK CIRCUIT GD151 (BK/GN) FOR AN OPEN

- Key in OFF position.



N0059540

Fig. 14: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between the sensor cluster C3037-5, circuit GD151 (BK/GN), harness side and ground.

- **Is the resistance less than 5 ohms?**

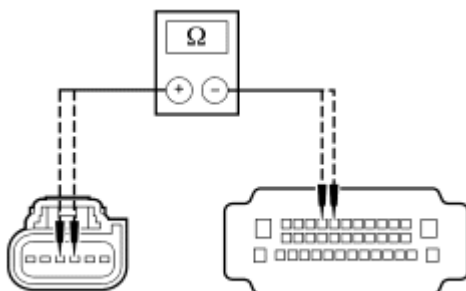
YES : Go to F5.

NO : REPAIR circuit GD151 (BK/GN). CLEAR the DTCs. RECALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. REPEAT the self-test.

F5 CHECK CIRCUITS VCA23 (BU/WH) AND VCA24 (GN/OG) FOR AN OPEN

NOTE: The battery and battery tray need to be removed to access the ABS module connector.

- Disconnect: ABS Module C135



N0059541

Fig. 15: Measuring Resistance
Courtesy of FORD MOTOR CO.

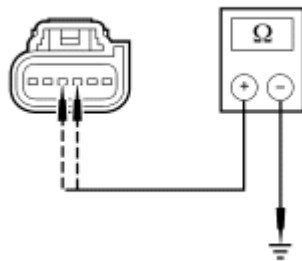
- Measure the resistance between:
 - ABS module C135-8, circuit VCA23 (BU/WH), harness side and stability control sensor cluster C3037-3, circuit VCA23 (BU/WH), harness side.
 - ABS module C135-9, circuit VCA24 (GN/OG), harness side and stability control sensor cluster C3037-4, circuit VCA24 (GN/OG), harness side.

- **Are the resistances less than 5 ohms?**

YES : Go to F6.

NO : REPAIR the affected circuit(s). CLEAR the DTCs. RECALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. REPEAT the self-test.

F6 CHECK CIRCUITS VCA23 (BU/WH) AND VCA24 (GN/OG) FOR A SHORT TO GROUND



N0059542

Fig. 16: Measuring Resistance Between Ground & Stability Control Sensor Cluster C3037-3 & C3037-4
Courtesy of FORD MOTOR CO.

- Measure the resistance between ground and:
 - stability control sensor cluster C3037-3, circuit VCA23 (BU/WH), harness side.
 - stability control sensor cluster C3037-4, circuit VCA24 (GN/OG), harness side.

- **Are the resistances greater than 10,000 ohms?**

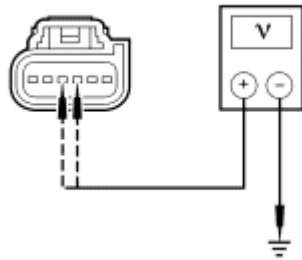
YES : Go to F7.

NO : REPAIR the affected circuit(s). CLEAR the DTCs. RECALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. REPEAT the self-test.

F7 CHECK CIRCUITS VCA23 (BU/WH) AND VCA24 (GN/OG) FOR A SHORT TO VOLTAGE

NOTE: The battery must be connected to carry out this test.

- Key in ON position.



N0059543

Fig. 17: Measuring Voltage Between Ground & Stability Control Sensor Cluster C3037-3 & C3037-4

Courtesy of FORD MOTOR CO.

- Measure the voltage between ground and:
 - stability control sensor cluster C3037-3, circuit VCA23 (BU/WH), harness side.
 - stability control sensor cluster C3037-4, circuit VCA24 (GN/OG), harness side.

- **Is any voltage present?**

YES : REPAIR the affected circuit(s). CLEAR the DTCs. RECALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. REPEAT the self-test.

NO : INSTALL a new sensor cluster. REFER to Stability Control Sensor Cluster. CLEAR the DTCs. RECALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. REPEAT the self-test.

Pinpoint Test G: DTC C1288 - Pressure Transducer Main/Primary Input Circuit Failure

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The ABS module monitors the brake pressure transducer in combination with other brake inputs; wheel speed sensors, stoplamp switch from the PCM over the High Speed Controller Area Network (HS-CAN) bus, lateral accelerometer from the sensor cluster. The pressure transducer signal is used by the ABS module for various diagnostic routines as well as to determine system functionality. The pressure transducer is located inside the Hydraulic Control Unit (HCU) and can be monitored via the master cylinder pressure (MC_PRESS) PID. The PID should correlate closely with system hydraulic pressure. When the vehicle is at rest without input from the brake pedal, this PID value should be at or close to 0 kPa (0 psi). However, due to factors such as vehicle age or minor discrepancies in manufacturing, PID readings may have some variability. If the transducer signal reads above 620 kPa (90 psi) when other inputs indicate the system should not be pressurized, DTC C1288 will be set. If the transducer requires service, a new HCU must be installed.

- DTC C1288 Pressure Transducer Main/Primary Input Circuit Failure - If during the ABS module self-test or normal operation the ABS module detects an open, a short to voltage or a short to ground in the transducer, DTC C1288 will be set. Also, if the transducer input does not match the information received from other sources, DTC C1288 will be set.

This pinpoint test is intended to diagnose the following:

- Brake pedal binding
- Air in the brake hydraulic system
- Fluid leak in the brake hydraulic system
- High-speed controller area network (HS-CAN) bus
- Brake on/off (BOO) switch
- HCU

PINPOINT TEST G: DTC C1288 - PRESSURE TRANSDUCER MAIN/PRIMARY INPUT CIRCUIT FAILURE

NOTE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

G1 CHECK FOR ACTIVE DTCs

- Connect the diagnostic tool.
- Key in ON position.
- Using the scan tool, clear the ABS module DTCs.
- Enter the following diagnostic mode on the diagnostic tool: Self-Test ABS Module
- Retrieve and record any ABS module DTCs.
- **Is DTC C1288 present?**

YES : Go to G2.

For all other ABS module DTCs, go to the **ABS MODULE DTC CHART** .

NO : The system is operating correctly at this time. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE and RECORD any DTCs.

G2 CHECK THE MASTER CYLINDER AND PEDAL FOR CORRECT OPERATION

- Press and release the brake pedal.
- **Does the brake pedal smoothly travel to the stop and then return automatically when released?**

YES : Go to G3.

NO : INSPECT the brake pedal and master cylinder for obstructions and foreign material that may cause a binding condition. REPAIR or INSTALL new as necessary. CLEAR the DTCs. REPEAT the self-test.

G3 CHECK FOR AIR IN THE BRAKE HYDRAULIC SYSTEM

- Select PARK.
- Start the engine.
- Press and release the brake pedal several times.
- **Does the brake pedal move smoothly and evenly through the length of it's travel?**

YES : Go to G4.

NO : REFER to **BRAKE SYSTEM - GENERAL INFORMATION** article to continue diagnosis of the brake hydraulic system.

G4 CHECK FOR FLUID LEAKS IN THE BRAKE HYDRAULIC SYSTEM

- Visually inspect the brake hydraulic system for brake fluid leaks.
- **Are any leaks found?**

YES : REPAIR as necessary. BLEED the brake system, REFER to **BRAKE SYSTEM - GENERAL INFORMATION** article. CLEAR the DTCs. REPEAT the self-test.

NO : Go to G5.

G5 MONITOR THE ABS MODULE BRAKE ON/OFF (BOO_ABS) PID

- Enter the following diagnostic mode on the diagnostic tool: DataLogger PCM
- Monitor the BOO_ABS PID while pressing and releasing the brake pedal.
- **Does the PID agree with the brake pedal position?**

YES : Go to G7.

NO : Go to G6.

G6 MONITOR THE PCM BRAKE ON/OFF (BOO) PID

- Enter the following diagnostic mode on the diagnostic tool: DataLogger PCM
- Monitor the BOO PID while pressing and releasing the brake pedal.
- **Does the PID agree with the brake pedal position?**

YES : REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose the HS-CAN bus.

NO : REFER to **EXTERIOR LIGHTING** article to continue diagnosis of the BOO switch.

G7 MONITOR THE ABS MODULE MASTER CYLINDER (MC_PRESS) PRESSURE PID

- Firmly press and release the brake pedal while monitoring the ABS module MC_PRESS PID.
- **Does the PID increase with increased brake pedal pressure and decrease with decreased brake pedal pressure?**

YES : The system is operating correctly at this time. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE and RECORD any DTCs.

NO : INSTALL a new HCU. REFER to **Hydraulic Control Unit (HCU)**. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

Pinpoint Test H: DTC C1446 - Brake Switch Circuit Failure

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Vehicle Dynamic Systems for schematic and connector information. Refer to appropriate SYSTEM WIRING DIAGRAMS article, Instrument Cluster for schematic and connector information.

Normal Operation

The ABS module monitors brake pedal position in combination with other brake inputs; wheel speed sensors,

brake pressure transducer, lateral accelerometer from the sensor cluster. The pedal position signal is used by the ABS module for various diagnostic routines as well as to determine system functionality. The Brake Pedal Position (BPP) switch is located inside the stoplamp switch and can be monitored via the brake on/off (BOO_ABS) PID and should match the actual brake pedal position. The ABS module receives the BPP switch information from the PCM over the High Speed Controller Area Network (HS-CAN) bus.

- **DTC C1446 Brake Switch Circuit Failure** - If the brake pedal switch status from the PCM is not received by the ABS module due to a failure of the BPP switch or the HS-CAN bus, the ABS module will set DTC C1446. The ABS module uses a pressure transducer inside the Hydraulic Control Unit (HCU) to verify the brake pedal switch status message from the PCM. When the vehicle is at rest without input from the brake pedal, the pressure transducer value should be at or close to 0 kPa (0 psi). However, due to factors such as vehicle age or minor discrepancies in manufacturing, readings may have some variability.

Possible Causes

- Wiring, terminals or connectors
- BPP switch
- HCU
- ABS module

PINPOINT TEST H: DTC C1446 - BRAKE SWITCH CIRCUIT FAILURE

H1 CHECK THE RED BRAKE WARNING INDICATOR

- Observe the red brake warning indicator while turning the ignition switch from OFF to RUN.
- Key in ON position.
- **Does the indicator illuminate for a short time and then go out?**

YES : Go to H2.

NO : If the red brake warning indicator remains illuminated or does not illuminate at all, go to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER & WARNING CHIMES** article to diagnose the red brake warning indicator.

H2 CHECK THE ABS MODULE MASTER CYLINDER PRESSURE (MC_PRES) PID

- Connect the diagnostic tool.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: DataLogger - ABS Module
- Monitor the MC_PRESS PID **without** pressing the brake pedal.
- **Does the PID display 0 kPa (0 psi)?**

YES : Go to H3.

NO : Go to **Pinpoint Test G** .

H3 CHECK FOR COMMUNICATION DTCs

- Review the DTCs retrieved and recorded during Inspection and Verification.
- **Are DTCs U0073 or U0115 present?**

YES : REFER to **MODULE COMMUNICATIONS NETWORK** article to diagnose the HS-

CAN bus.

NO : Go to H4.

H4 MONITOR THE ABS MODULE BRAKE ON/OFF (BOO) PID

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - ABS Module
- Monitor the BOO PID, while pressing and releasing the brake pedal.
- **Does the PID agree with the brake pedal position?**

YES : The system is operating correctly at this time. TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. RETRIEVE and RECORD any DTCs.

NO : Go to H5.

H5 MONITOR THE PCM BRAKE PEDAL POSITION (BOO) PID

- Enter the following diagnostic mode on the diagnostic tool: DataLogger - PCM
- Monitor the BOO PID, while pressing and releasing the brake pedal.
- **Does the PID agree with the brake pedal position?**

YES : REFER to **MODULE COMMUNICATIONS NETWORK** article to continue diagnosis of the HS-CAN bus.

NO : REFER to **EXTERIOR LIGHTING** article to continue diagnosis of the stoplamps.

Pinpoint Test I: DTC C1991 - Module Calibration Failure

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The ABS module must be calibrated whenever a component specific to the roll stability control (RSC) system is disconnected or a new component is installed.

- DTC C1991 Module Calibration Failure - If the calibration procedure does not finish or if the ABS module loses the calibration information, DTC C1991 will be set.

This pinpoint test is intended to diagnose the following:

- ABS module
- Stability control sensor cluster
- Steering angle sensor module (SASM)

PINPOINT TEST I: DTC C1991 - MODULE CALIBRATION FAILURE

NOTE: **The RSC indicator flashes when a system calibration is required. Calibration is required after certain DTCs are cleared.**

NOTE: **Make sure the vehicle is not on a hoist and is sitting on level ground before**

carrying out ABS module calibration.**I1 CHECK FOR DTCs**

- Connect the diagnostic tool.
- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: Self-Test ABS Module
- Retrieve and record any ABS module DTCs.
- **Are other DTCs retrieved along with C1991?**

YES : Go to the **ABS MODULE DTC CHART** .

Some of the pinpoint tests recommend ABS module calibration prior to any further diagnostics. In this case, go to I2.

NO : Go to I2.

I2 CALIBRATE THE ABS MODULE

- Enter the following diagnostic mode on the diagnostic tool: ABS Module Calibration
- Carry out the ABS module calibration per the scan tool instructions.
- **Does the calibration complete for all of the RSC sensors, and does the RSC indicator stop flashing and turn off?**

YES : Go to I3.

NO : Go to I4.

Some concerns cause the calibration not to complete. If sent here from another pinpoint test, return to that pinpoint test for further diagnostics.

I3 CLEAR DTC C1991

- Enter the following diagnostic mode on the diagnostic tool: Clear ABS Module DTCs
- Using the scan tool, clear the DTC.
- **Does the DTC clear?**

YES : TEST DRIVE the vehicle above 32 km/h (20 mph) and CARRY OUT at least one ABS stop. REPEAT the self-test.

If sent here from another pinpoint test, return to that pinpoint test for further diagnostics.

NO : Go to I4.

I4 MONITOR THE ABS MODULE SASM (SWA POS) PID

- Key in ON position.
- Enter the following diagnostic mode on the diagnostic tool: DataLogger ABS Module
- Monitor the SWA POS PID while rotating the steering wheel clockwise and counterclockwise very slowly.
- **Does the PID move in intervals of 1 degree?**

YES : Go to **Pinpoint Test F** .

NO : Go to **Pinpoint Test E** .

Pinpoint Test J: DTC U1901 - CAN Network #2 Communication Bus Fault - Receive Error

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The ABS module communicates with the stability control sensor cluster along the secondary high-speed controller area network (HS-CAN2) bus, circuits VCA23 (BU/WH) and VCA24 (GN/OG).

- DTC U1901 CAN Network #2 Communication Bus Fault - Receive Error - If the ABS module does not receive an expected message from the stability control sensor cluster or the message received does not agree with other sensor inputs, DTC U1901 will be set.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Stability control sensor cluster
- ABS module

PINPOINT TEST J: DTC U1901 - CAN NETWORK #2 COMMUNICATION BUS FAULT- RECEIVE ERROR

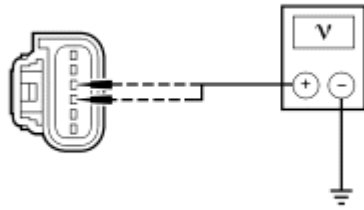
NOTE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

J1 CHECK CIRCUITS VCA23 (BU/WH) AND VCA24 (GN/OG) FOR A SHORT TO VOLTAGE

NOTE: The battery and battery tray need to be removed to access the ABS module connector.

NOTE: The battery must be connected to carry out this test.

- Key in OFF position.
- Disconnect: ABS Module C135
- Disconnect: Stability Control Sensor Cluster C3037
- Key in ON position.



N0034609

Fig. 18: Checking Circuits For Voltage
Courtesy of FORD MOTOR CO.

- Measure the voltage between ground and:
 - stability control sensor cluster C3037-3, circuit VCA23 (BU/WH), harness side.
 - stability control sensor cluster C3037-4, circuit VCA24 (GN/OG), harness side.

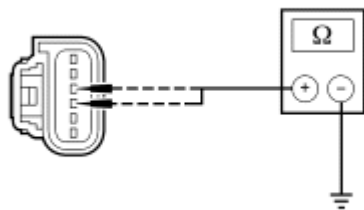
Is any voltage present?

YES : Go to J2.

NO : REPAIR the affected circuit(s). CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

J2 CHECK CIRCUITS VCA23 (BU/WH) AND VCA24 (GN/OG) FOR A SHORT TO GROUND

- Key in OFF position.



N0034608

Fig. 19: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between ground and:
 - stability control sensor cluster C3037-3, circuit VCA23 (BU/WH), harness side.
 - stability control sensor cluster C3037-4, circuit VCA24 (GN/OG), harness side.

Is the resistances greater than 10,000 ohms?

YES : Go to J3.

NO : REPAIR the affected circuit(s). CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

J3 CHECK CIRCUITS VCA23 (BU/WH) AND VCA24 (GN/OG) FOR AN OPEN

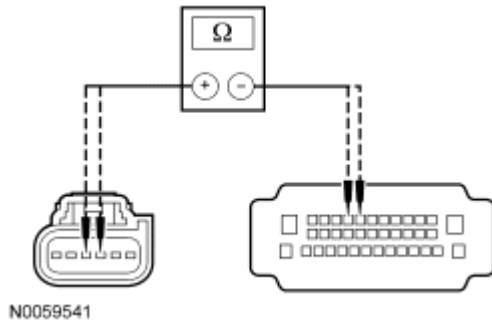


Fig. 20: Measuring Resistance
Courtesy of FORD MOTOR CO.

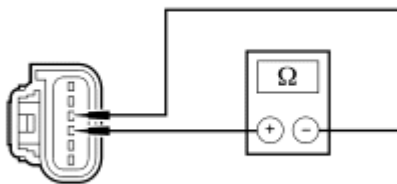
- Measure the resistance between:
 - stability control sensor cluster C3037-3, circuit VCA23 (BU/WH), harness side and ABS module C135-8, circuit VCA23 (BU/WH), harness side.
 - stability control sensor cluster C3037-4, circuit VCA24 (GN/OG), harness side and ABS module C135-9, circuit VCA24 (GN/OG), harness side.

• **Are the resistances less than 5 ohms?**

YES : Go to J4.

NO : REPAIR the affected circuit(s). CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

J4 CHECK CIRCUITS VCA23 (BU/WH) AND VCA24 (GN/OG) FOR A SHORT TOGETHER



N0034610

Fig. 21: Measuring Resistance
Courtesy of FORD MOTOR CO.

- Measure the resistance between stability control sensor cluster C3037-3, circuit VCA23 (BU/WH), harness side and stability control sensor cluster C3037-4, circuit VCA24 (GN/OG), harness side.

• **Is the resistance greater than 10,000 ohms?**

YES : Go to J5.

NO : REPAIR the affected circuit(s). CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

J5 CHECK THE SENSOR CLUSTER MODULE

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

- Install a known good sensor cluster. Refer to **Stability Control Sensor Cluster**.
- Connect: ABS Module C135
- Carry out the ABS module self-test. Retrieve and record any ABS module DTCs.
- **Was DTC U1901 retrieved?**

YES : INSTALL the original sensor cluster. INSTALL a new ABS module. REFER to **Anti-Lock Brake System (ABS) Module**. CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied

NO : Fault corrected. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

Pinpoint Test K: The Traction Control System Cannot Be Disabled

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Instrument Cluster for schematic and connector information.

Normal Operation

The traction control switch is hard wired to the instrument cluster, traction control system status is indicated by an indicator in the instrument cluster. When the driver deactivates the traction assist system, the instrument cluster sends a message to the PCM and the ABS module over the high-speed controller area network (HS-CAN) bus. The traction assist system will remain deactivated until the driver pushes the switch again or until the ignition switch is cycled.

The instrument cluster sends a reference voltage to the stability/traction control switch along circuit CCA15 (YE/GY). When the switch is pressed, the reference voltage is momentarily grounded through the switch along circuit GD116 (BK/VT).

- DTC C1093 Traction Control Disable Switch Circuit Failure - DTC C1093 will be set if the stability/traction control switch is pressed for more than 2 minutes during normal operation, is pressed at all during the instrument cluster on-demand self test or if there is an open circuit, a short to voltage or a short.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Traction control switch
- Instrument cluster

PINPOINT TEST K: THE TRACTION CONTROL SYSTEM CANNOT BE DISABLED

NOTE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

K1 CHECK CIRCUIT CCA15 (YE/GY) FOR VOLTAGE

- Disconnect: Traction Control Switch C2355
- Key in ON position.

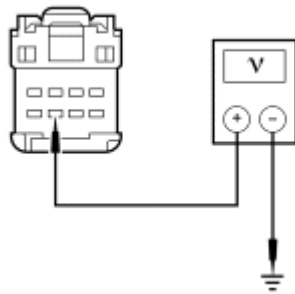


Fig. 22: Measuring Voltage Between Stability/Traction Control Switch C202-7, Circuit CCA15 (YE/GY)
Courtesy of FORD MOTOR CO.

- Measure the voltage between stability/traction control switch C2355-7, circuit CCA15 (YE/GY), harness side.
- **Is the voltage greater than 9.5 volts?**
YES : Go to K2.
NO : Go to K4.

K2 CHECK CIRCUIT GD116 (BK/VT) FOR AN OPEN

- Key in OFF position.

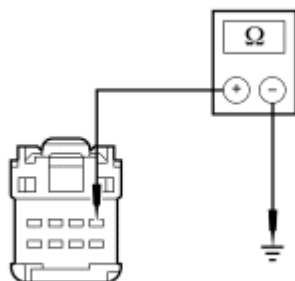


Fig. 23: Measuring Resistance Between Stability/Traction Control Switch C202-1, Circuit GD116 (BK/VT)
Courtesy of FORD MOTOR CO.

- Measure the resistance between stability/traction control switch C2355-1, circuit GD116 (BK/VT), harness side and ground.
- **Is the resistance less than 5 ohms?**
YES : Go to K3.
NO : REPAIR circuit GD116 (BK/VT). CLEAR the DTCs. TEST the system for normal operation.

K3 ISOLATE THE STABILITY/TRACTION CONTROL SWITCH

- Measure the resistance between stability/traction control switch C2355 pin-7, component side and pin-1, component side while pressing and releasing the switch.
- **Is the resistance less than 5 ohms with the switch pressed and greater than 10,000 ohms with the switch released?**

YES : Go to K6.

NO : INSTALL a new stability/traction control switch. REFER to Stability/Traction Control Switch. CLEAR the DTCs. TEST the system for normal operation.

K4 CHECK CIRCUIT CCA15 (YE/GY) FOR AN OPEN

- Key in OFF position.
- Disconnect: Instrument Cluster C220

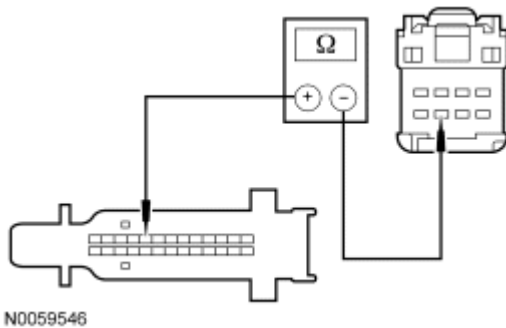


Fig. 24: Measuring Resistance Between Stability/Traction Control Switch C202-7, CCA15 (YE/GY) & Instrument Cluster C220-22
Courtesy of FORD MOTOR CO.

- Measure the resistance between stability/traction control switch C2355-7, CCA15 (YE/GY), harness side and instrument cluster C220-22, circuit CCA15 (YE/GY), harness side.

- **Is the resistance less than 5 ohms?**

YES : Go to K5.

NO : REPAIR circuit CCA15 (YE/GY). CLEAR the DTCs. TEST the system for normal operation.

K5 CHECK CIRCUIT CCA15 (YE/GY) FOR A SHORT TO VOLTAGE

- Key in ON position.

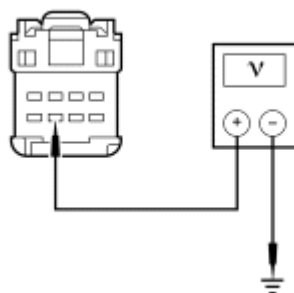


Fig. 25: Measuring Voltage Between Stability/Traction Control Switch C202-7, CCA15 (YE/GY) & Ground**Courtesy of FORD MOTOR CO.**

- Measure the voltage between stability/traction control switch C2355-7, CCA15 (YE/GY), harness side and ground.

- **Is any voltage present?**

YES : REPAIR circuit CCA15 (YE/GY). CLEAR the DTCs. TEST the system for normal operation.

NO : Go to K6.

K6 CHECK THE INSTRUMENT CLUSTER ELECTRICAL CONNECTOR

- Check instrument cluster C220 for:
 - corrosion.
 - pushed-out pins.
 - spread terminals.
- Connect: Instrument Cluster C220
- Make sure the instrument cluster connector seats correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new instrument cluster. REFER to **INSTRUMENT CLUSTER (IC), MESSAGE CENTER & WARNING CHIMES** article. CLEAR the DTCs. TEST the system for normal operation.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

Pinpoint Test L: DTCs B1317 and B1318 - Battery Voltage High/Low

Refer to appropriate SYSTEM WIRING DIAGRAMS article, Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The voltage range required to operate the steering angle sensor module (SASM) is 8-16 volts.

Fused ignition voltage is supplied to the SASM from smart junction box (SJB) fuse 34 (5A) along circuit CBP34 (VT/BN), ground is provided along circuit GD115 (BK/GY). The module communicates with other modules along the high-speed controller area network (HS-CAN) bus, circuits VDB05 (WH) and VDB04 (WH/BU).

- DTC B1317 Battery Voltage High - If the battery voltage is higher than 16 volts for more than 100 ms, DTC B1317 will be set.
- DTC B1318 Battery Voltage Low - If the battery voltage is lower than 8 volts for more than 100 ms,

DTC B1317 will be set.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- SASM
- Charging system

PINPOINT TEST L: DTCs B1317 AND B1318 - BATTERY VOLTAGE HIGH/LOW

NOTE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

L1 CHECK FOR RETURNING DTCs

- Connect the diagnostic tool.
- Key in ON position.
- Carry out the SASM self-test. Retrieve and record all SASM DTCs.
- **Are DTCs B1317 or 1318 present?**

YES : Go to L2.

NO : REFER to **CHARGING SYSTEM - GENERAL INFORMATION** article to continue diagnosis of the charging system.

L2 CHECK THE BATTERY VOLTAGE

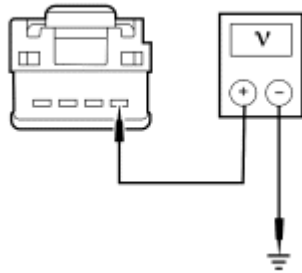
- Measure the voltage between the positive and negative battery terminals with the key ON and the engine OFF (KOEO) and with the engine running.
- **Is the battery voltage between 10 and 13 volts with KOEO and between 13 and 17 volts with the engine running?**

YES : Go to L3.

NO : REFER to **CHARGING SYSTEM - GENERAL INFORMATION** article to continue diagnosis of the charging system.

L3 CHECK THE VOLTAGE TO THE SASM

- Key in OFF position.
- Disconnect: SASM C222
- Key in ON position.



N0059533

Fig. 26: Measuring Voltage Between Steering Angle Sensor C222-1, CBP34 (VT/BN), Harness Side & Ground

Courtesy of FORD MOTOR CO.

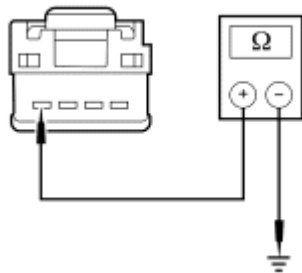
- Measure the voltage between SASM C222-1, CBP34 (VT/BN), harness side and ground.
- **Is the voltage greater 10 volts?**

YES : Go to L4.

NO : VERIFY SJB fuses 34 (5A) is OK. If OK, REPAIR circuit CBP34 (VT/BN). CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

L4 CHECK THE CIRCUIT GD115 (BK/GY) FOR AN OPEN

- Key in OFF position.



N0059535

Fig. 27: Measuring Resistance Between Steering Angle Sensor C222-4, GD115 (BK/GY), Harness Side & Ground

Courtesy of FORD MOTOR CO.

- Measure the resistance between SASM C222-4, GD115 (BK/GY), harness side and ground.
- **Is the resistance less than 5 ohms?**

YES : Go to L5.

NO : REPAIR circuit GD115 (BK/GY). CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure. CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

L5 CHECK THE ELECTRICAL CONNECTOR SASM

- Check SASM C222 for:

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

- corrosion.
- pushed-out pins.
- spread terminals.
- Connect: SASM C222
- Make sure the ABS connector seats correctly.
- Operate the system and verify the concern is still present.
- **Is the concern still present?**

YES : INSTALL a new SASM. REFER to **Steering Angle Sensor Module (SASM)**.

CALIBRATE the ABS module. FOLLOW the scan tool directions for the calibration procedure.
CLEAR the DTCs. CARRY OUT the self-test with the brake pedal not applied.

NO : The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test.

REMOVAL AND INSTALLATION

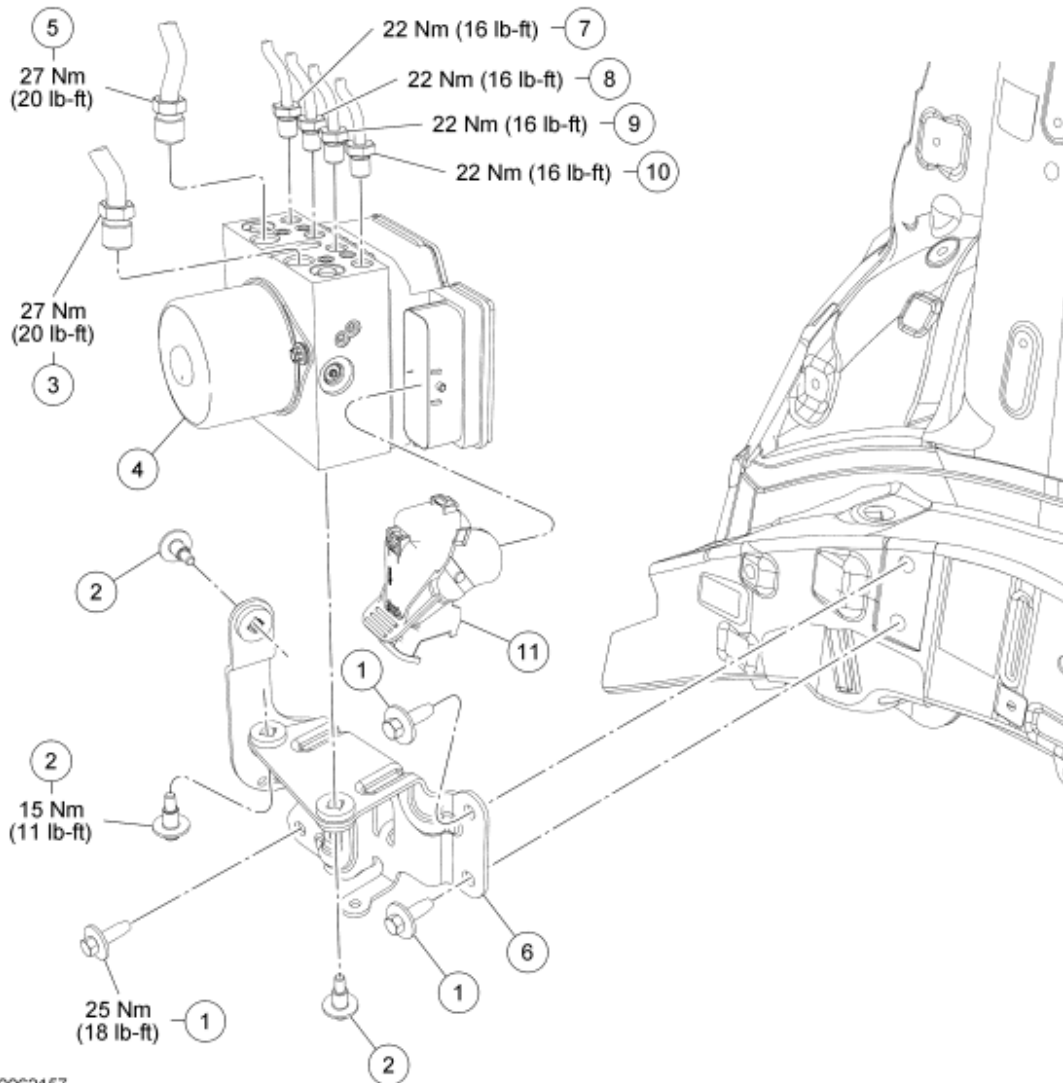
HYDRAULIC CONTROL UNIT (HCU)

Material

Item	Specification
High Performance DOT 3 Motor Vehicle Brake Fluid PM-1-C (US); CPM-1-C (Canada)	WSS-M6C62-A or WSS-M6C65-A1

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX



N0062157

Fig. 28: Exploded View Of Hydraulic Control Unit (HCU) With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W707402	Hydraulic control unit (HCU) bracket-to-frame bolts (3 required)
2	-	HCU bracket-to-HCU bolts (3 required) (part of 2C286)
3	2286	Master cylinder rear brake tube
4	2C286	HCU
5	2A335	Master cylinder front brake tube
6	2C304	HCU bracket
7	2263	RF brake caliper brake tube
8	2C293	RR brake caliper brake tube

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

9	2C294	LR brake caliper brake tube
10	2264	LF brake caliper brake tube
11	-	ABS module electrical connector (part of 14290)

REMOVAL AND INSTALLATION

WARNING: Do not use any fluid other than clean brake fluid meeting manufacturer's specification. Additionally, do not use brake fluid that has been previously drained. Following these instructions will help prevent system contamination, brake component damage and the risk of serious personal injury.

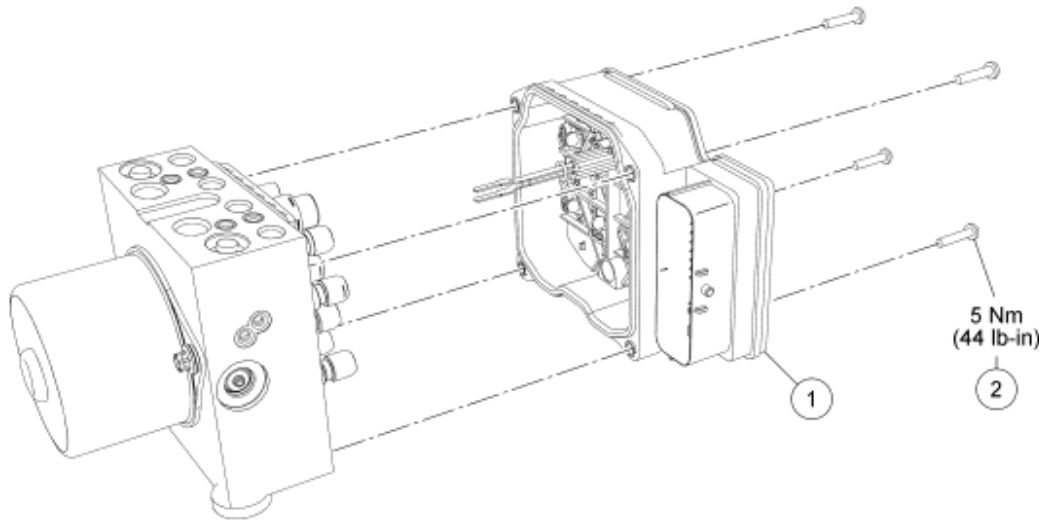
WARNING: Carefully read cautionary information on product label. For EMERGENCY MEDICAL INFORMATION seek medical advice. In the USA or Canada on Ford/Motorcraft products call: 1-800-959-3673. For additional information, consult the product Material Safety Data Sheet (MSDS) if available. Failure to follow these instructions may result in serious personal injury.

NOTE: Brake fluid is harmful to painted and plastic surfaces. If brake fluid is spilled onto a painted or plastic surface, immediately wash it with water.

1. Remove the air cleaner outlet tube and the air cleaner assembly. For additional information, refer to **INTAKE AIR DISTRIBUTION AND FILTERING** article.
2. Remove the battery and battery tray. For additional information, refer to **BATTERY, MOUNTING & CABLES** article.
3. Disconnect the ABS module electrical connector.
4. Disconnect the 6 brake tube fittings.
 - To install, tighten the brake master cylinder tubes to 27 Nm (20 lb-ft).
 - To install, tighten the remaining tubes to 22 Nm (16 lb-ft).
5. Remove the 3 hydraulic control unit (HCU) bracket-to-frame bolts and the HCU.
 - To install, tighten to 25 Nm (18 lb-ft).
6. Remove the 3 HCU bracket-to-HCU bolts and separate the HCU from the bracket.
 - To install, tighten to 15 Nm (11 lb-ft).

NOTE: Make sure the HCU and ABS module are clean and free of any brake fluid or foreign material before separating the components.

7. To install, reverse the removal procedure.
 - Bleed the brake system. For additional information, refer to **BRAKE SYSTEM - GENERAL INFORMATION** article.

ANTI-LOCK BRAKE SYSTEM (ABS) MODULE

N0059464

Fig. 29: Exploded View Of Anti-Lock Brake System (ABS) Module With Torque Specification
 Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	2C219	ABS module
2	-	ABS module screw (4 required) (part of 2C219)

REMOVAL AND INSTALLATION

NOTE: Electronic modules are sensitive to electrical charges. The anti-lock brake system (ABS) module can be damaged if exposed to these charges.

NOTE: Do not allow any brake fluid or foreign material to enter the mating side of the anti-lock brake system (ABS) module or damage to the solenoids can occur.

1. Remove the hydraulic control unit (HCU). For additional information, refer to **Hydraulic Control Unit (HCU)**.
2. Remove the 4 ABS module screws and the ABS module.
 - To install, tighten to 5 Nm (44 lb-in).

NOTE: Visually inspect the ABS module terminals for damage before installation.

3. To install, reverse the removal procedure.

WHEEL SPEED SENSOR - FRONT

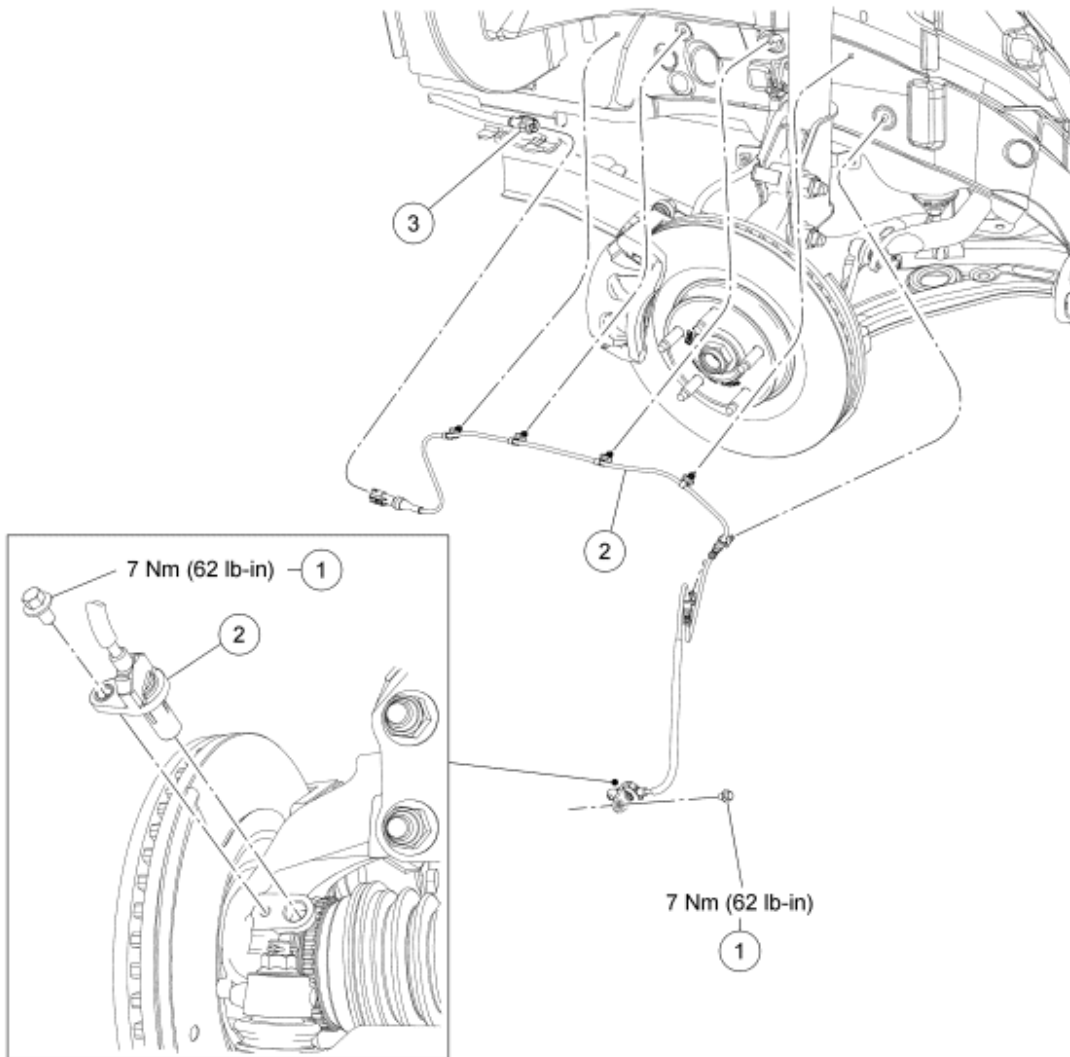


Fig. 30: Exploded View Of Front Wheel Speed Sensor With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W500010	Wheel speed sensor bolt
2	2C204 RH/2C205 LH	Wheel speed sensor
3	-	Wheel speed sensor electrical connector (part of 14290)

REMOVAL AND INSTALLATION

1. Remove the wheel and tire. For additional information, refer to **WHEELS & TIRES** article.
2. Remove the retainers and position the fender splash shield aside.
3. Disconnect the wheel speed sensor electrical connector.
4. Disconnect the 6 pushpin fasteners.
5. Remove the front wheel speed sensor bolt and the wheel speed sensor.
 - To install, tighten to 7 Nm (62 lb-in).

NOTE: **The wheel speed sensor harness must be routed as shown or damage to the harness during vehicle jounce and rebound can occur.**

6. To install, reverse the removal procedure.

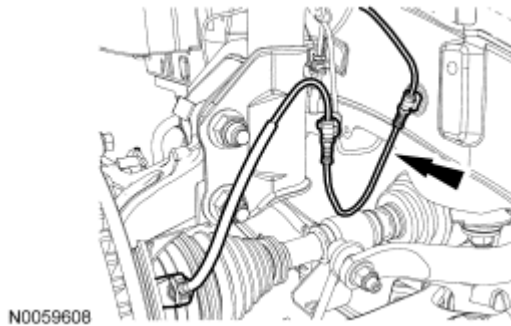


Fig. 31: Identifying Wheel Speed Sensor Harness
Courtesy of FORD MOTOR CO.

WHEEL SPEED SENSOR - REAR

NOTE: **All wheel drive (AWD) shown, front wheel drive (FWD) similar.**

2008 Ford Edge SE

2008 BRAKES Vehicle Dynamic Systems - Edge & MKX

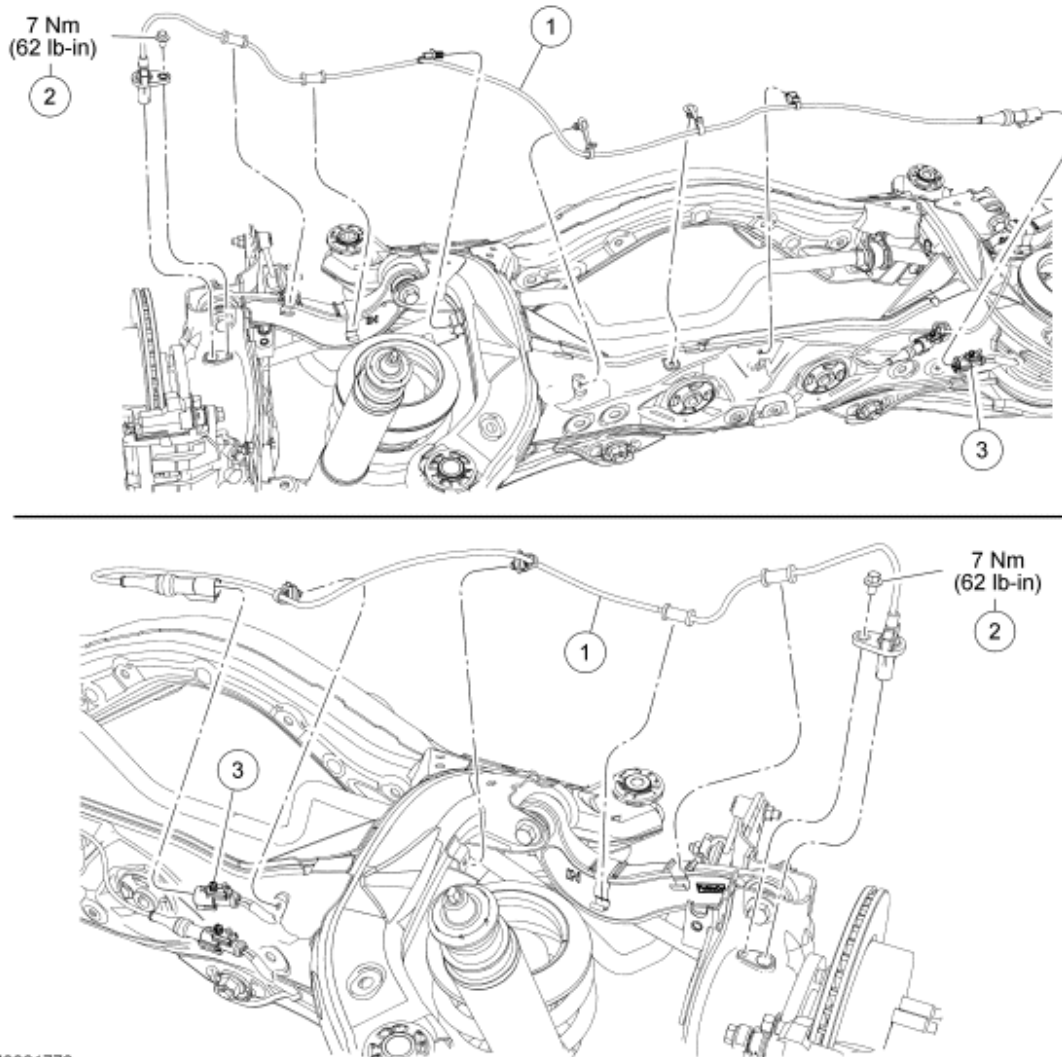


Fig. 32: Exploded View Of Rear Wheel Speed Sensor With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	2C190	Wheel speed sensor
2	W500010	Wheel speed sensor bolt
3	-	Wheel speed sensor electrical connector (part of 14A005)

REMOVAL AND INSTALLATION

1. Remove the wheel and tire. For additional information, refer to **WHEELS & TIRES** article.
2. Disconnect the wheel speed sensor electrical connector.

NOTE: It is not necessary to remove the harness routing brackets.

3. Disconnect the wheel speed sensor harness from the brackets.
4. Disconnect the pushpin fasteners.
5. Remove the wheel speed sensor bolt and the wheel speed sensor.
 - To install, tighten to 7 Nm (62 lb-in).
6. To install, reverse the removal procedure.

STABILITY CONTROL SENSOR CLUSTER

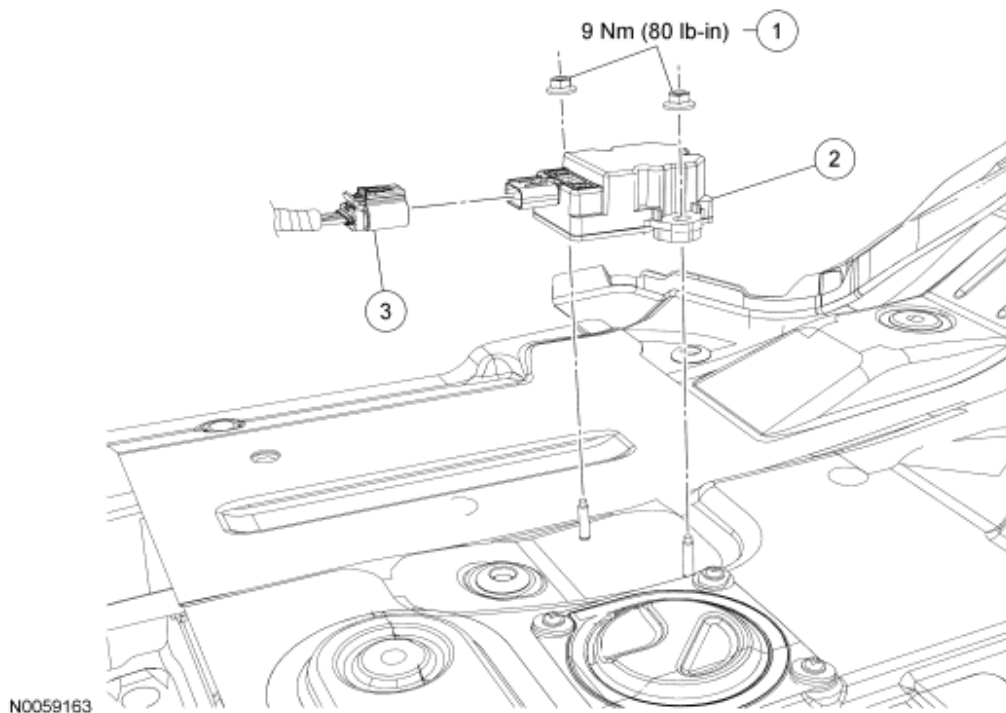


Fig. 33: Exploded View Of Stability Control Sensor Cluster With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W707142	Stability control sensor cluster nuts (2 required)
2	3C187	Stability control sensor cluster
3	-	Stability control sensor cluster electrical connector (part of 14A005)

REMOVAL AND INSTALLATION

NOTE: When installing a new stability control sensor cluster, the sensor cluster must be calibrated. Follow the scan tool directions for the calibration procedures.

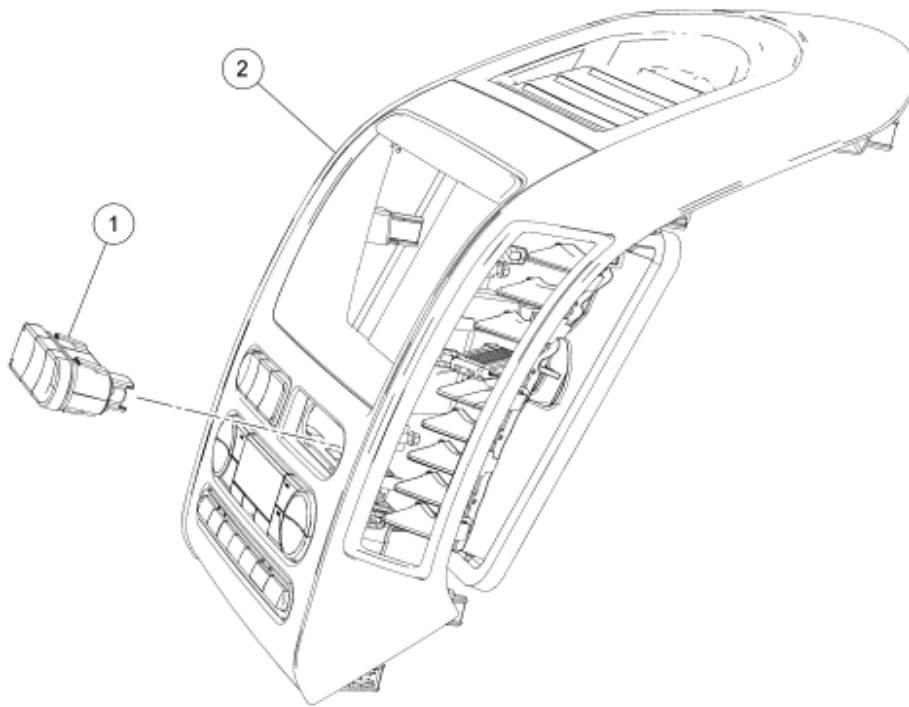
NOTE: The sensor is located on the RH side of the vehicle, under the rear seat.

1. Remove the rear seat. For additional information, refer to **SEATING** article.
2. Remove the right rear scuff plate. For additional information, refer to **INTERIOR TRIM & ORNAMENTATION** article.
 - Position the carpet aside.
3. Disconnect the stability control sensor cluster electrical connector.
4. Remove the 2 stability control sensor cluster nuts and the stability control sensor cluster.
 - To install, tighten to 9 Nm (80 lb-in).
5. To install, reverse the removal procedure.

STABILITY/TRACTION CONTROL SWITCH

REMOVAL AND INSTALLATION

NOTE: Edge shown, MKX similar.



N0085330

Fig. 34: Identifying Stability/Traction Control Switch
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	13D730	Stability/traction control switch
2	7804302	Instrument panel center finish panel

NOTE: The ignition switch must be in the OFF position if disconnecting the Passenger Air Bag Deactivation (PAD) indicator electrical connector. Otherwise, a DTC will be set in the Restraints Control Module (RCM) memory and must be cleared before releasing the vehicle.

- 1. Turn the ignition OFF and wait at least one minute.
- 2. Remove the instrument panel center finish panel. For additional information, refer to **INSTRUMENT PANEL & CONSOLE** article.
- 3. Depress the 4 tabs and remove the stability/traction control switch from the instrument panel center finish panel.

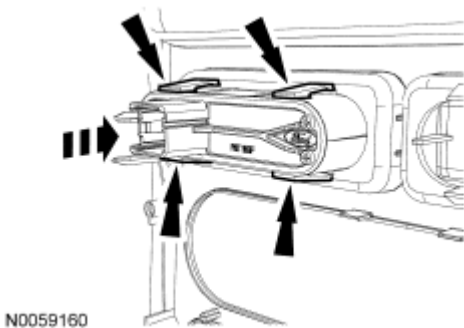


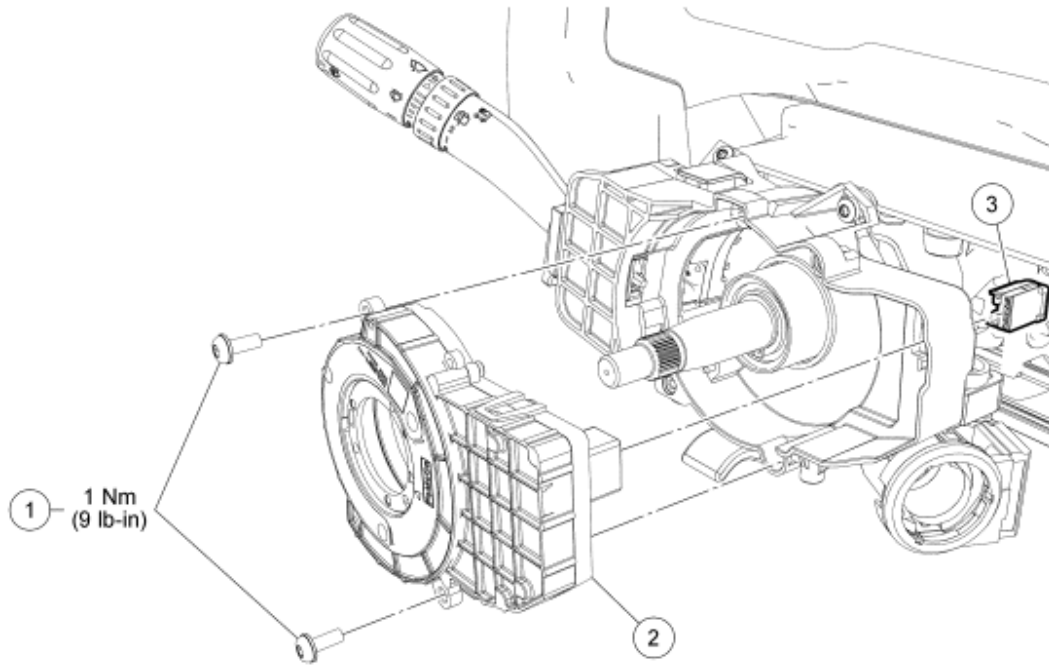
Fig. 35: Depressing Tabs & Removing Stability/Traction Control Switch From Instrument Panel Center Finish Panel
Courtesy of FORD MOTOR CO.

- 4. To install, reverse the removal procedure.

STEERING ANGLE SENSOR MODULE (SASM)

Special Tools

Illustration	Tool Name	Tool Number
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool	



N0061774

Fig. 36: Exploded View Of Steering Wheel Rotation Sensor With Torque Specification
 Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Steering angle sensor module (SASM) bolts (2 required) (part of 13D774)
2	3F818	SASM
3	-	SASM electrical connector (part of 14401)

REMOVAL

1. Remove the clockspring. For additional information, refer to **SUPPLEMENTAL RESTRAINT SYSTEM** article.
2. Remove the 2 steering angle sensor module (SASM) bolts.
3. Disconnect the SASM electrical connector and remove the steering wheel rotation sensor.

INSTALLATION

1. If the arrow on the SASM ring and the arrow on the sensor housing are misaligned and/or the straight ahead position of the sensor is not known, follow these steps to align the sensor prior to installing the sensor:
 1. Orient the front wheels in the straight-ahead position.
 2. Connect the SASM electrical connector.

3. Connect the battery negative cable. For additional information, refer to **BATTERY, MOUNTING & CABLES** article.
4. Carefully rotate the sensor clockwise until it stops.
5. Rotate the sensor counter-clockwise 2-1/2 turns.
6. Using the scan tool, read the SASM position PID (SWA POS). The PID should read approximately 0 degrees. If not, repeat the procedure.
7. Disconnect the battery negative cable. For additional information, refer to **BATTERY, MOUNTING & CABLES** article.
8. Disconnect the SASM electrical connector.

NOTE: Slight rotation of the SASM ring is allowed to align the 2 arrows.

2. Make sure the sensor ring arrow is lined up with the SASM housing arrow as shown and position the sensor into the multi-function switch housing.

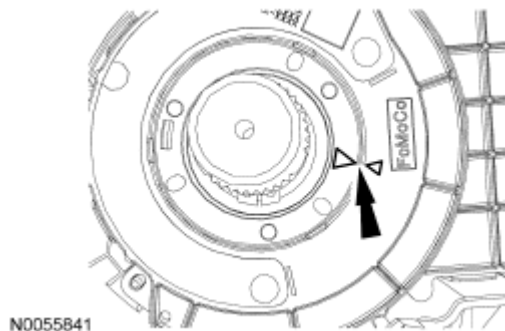


Fig. 37: Making Sure Sensor Ring Arrow Is Lined Up With Absolute Steering Angle Sensor Housing Arrow

Courtesy of FORD MOTOR CO.

3. Connect the SASM electrical connector.
4. Remove the 2 SASM bolts.
 - Tighten to 1 Nm (9 lb-in).
5. Install the clockspring. For additional information, refer to **SUPPLEMENTAL RESTRAINT SYSTEM** article.
 - If the SASM was aligned using the scan tool, DTCs will be present in the restrains control module (RCM). Using the scan tool, clear the DTCs in the RCM.